

Talking Abacws

Emergency Responders

A decision-focused stakeholder question catalogue

80 practical questions for evidence-bounded building information before arrival, during incident operations, through command handover and into recovery.

80

QUESTIONS

8

CATEGORIES

6

FIELDS PER CARD

EVIDENCE PRECEDENCE

- 1 Incident command, emergency-service information and direct observation
- 2 Direct authorised observations and calibrated responder instruments
- 3 Current competent site owners and authoritative system panels
- 4 Commissioned, healthy and permissioned telemetry
- 5 Current controlled records and verified as-built information
- 6 May 2021 floor plan - historical orientation only

All sensing, integrations and analytics are proposed until deployment, mapping, commissioning, permission and current health are independently verified.

Scope and safe-use statement

The service organises evidence for responders; it cannot replace incident command, direct observation or competent judgement.

PURPOSE

1 PURPOSE

Support faster orientation to current site information before arrival and during handover.

2 PURPOSE

Expose source, owner, time, permission, provenance, uncertainty and conflict for every answer.

3 PURPOSE

Return explicit unavailable or conflicting states instead of filling evidence gaps with plausible text.

4 PURPOSE

Connect operational evidence, preservation needs and recovery dependencies to accountable owners.

OPERATIONAL BOUNDARIES

1 OPERATIONAL BOUNDARIES

No operational commands, system control, isolation, authorisation or certification.

2 OPERATIONAL BOUNDARIES

No prediction of individual survival, behaviour, exposure, mobility or evacuation outcome.

3 OPERATIONAL BOUNDARIES

No person identification, facial recognition, personal-device tracking or movement histories.

4 OPERATIONAL BOUNDARIES

No experimental telemetry may override incident command, emergency-service information or direct observation.

EVIDENCE STATUS

1 EVIDENCE STATUS

Sensors, integrations and analytics remain proposed until independently deployed, mapped, commissioned and permissioned.

2 EVIDENCE STATUS

Current authoritative owners and competent professionals always take precedence.

3 EVIDENCE STATUS

The May 2021 floor plan is historical orientation only and cannot prove current conditions.

4 EVIDENCE STATUS

Every answer exposes time basis, source health, gaps, drift, permissions, provenance and unresolved conflicts.

Required safe failure: when evidence, permission, spatial coverage, time alignment, calibration, mapping or system health is insufficient, return an explicit unavailable, conflicting or unconfirmed state and identify the evidence or owner needed next.

Field guide for every question card

Each field keeps a responder decision tied to evidence quality, authority, analysis limits and safe failure.

1

STAKEHOLDER QUESTION

A natural, role-specific question for mobilising control, first-arriving crews, incident command, specialist teams or recovery leads.

2

SENSORS AND TELEMETRY

Proposed modality and installation point, spatial coverage, event or sample interval, aggregation, privacy, calibration or functional test, clocks, uptime, gaps, drift, provenance and physical limits.

3

AUTHORITATIVE SUPPORTING DATA

Current owners and source precedence, effective, observed, source-event and retrieved times, permissions, version, provenance, staleness and conflicts.

4

ANALYSIS REQUIRED

Operation type, decision interval, minimum evidence, method, uncertainty, estimate or recommendation boundary and explicit safe failure.

5

ANSWER BOUNDARY

What the service must not infer, disclose, promise, certify, authorise, operate or control.

6

WHY THIS IS USEFUL

The concrete responder decision, handover, resource, access, evidence or recovery action supported by the answer.

Evidence, measurement and analysis discipline

A useful answer shows what was measured or looked up, when it applied, how it was derived and why it may still be wrong.

MEASUREMENT ASSURANCE

MODALITY AND LOCATION

Name the device or record, physical asset, installation point and mapping version.

COVERAGE AND INTERVAL

State monitored boundaries, unmonitored gaps, event or sample interval and aggregation window.

CALIBRATION OR TEST

Use calibration for measured variables and functional or end-to-end tests for binary states.

CLOCKS AND HEALTH

Show source clock offset, latency, uptime, gaps, saturation, resets and drift.

PROVENANCE

Retain raw event or sample identifiers, units, firmware, transformations and retrieval path.

FOUR TIMES TO DISPLAY

EFFECTIVE TIME

When a plan, role, restriction or temporary control is intended to apply.

OBSERVED TIME

When an authorised person or calibrated instrument observed the condition.

SOURCE-EVENT TIME

When the originating panel, controller or device recorded the event.

RETRIEVED TIME

When the responder-facing service obtained or refreshed the information.

CLOCK QUALITY

Display offset, synchronisation source and uncertainty; never hide unknown alignment.

OPERATION TYPES

LOOKUP / OBSERVATION

Retrieve a current record or show a direct, instrument or authorised observation.

CALCULATION / ESTIMATE

Retain inputs and units; label assumptions, range and error.

COMPARISON / DIAGNOSIS

Keep conflicts visible; do not claim root cause without competent confirmation.

FORECAST

Use only with a verified model, horizon and error bound; otherwise show current trend only.

RECOMMENDATION

Present bounded options for command assessment, never an instruction.

PERMITTED ANSWER STATES

CONFIRMED

Current authoritative or direct evidence meets the stated minimum.

REPORTED

An accountable source reports the condition; direct confirmation is absent.

ESTIMATED

A bounded aggregate or model is used and assumptions are visible.

CONFLICTING

Current sources disagree and precedence does not resolve the conflict.

UNAVAILABLE

Evidence, permission, mapping, coverage, timing or health is insufficient.

STAKEHOLDER QUESTION

ER-001

Which verified premises, external responder entrance and address reference should mobilising control pass to the first-arriving appliance?

SENSORS AND TELEMETRY

Proposed evidence. Proposed alarm and building-event interfaces can attach a stable site identifier, alarm panel address and mapped entrance reference to the incident record; entrance contacts may report open, closed or faulted state. Coverage. Every Abacws building identifier, street address, external responder entrance and panel-to-building mapping. Timing, aggregation and privacy. Event-driven at alarm creation and change, with a 60-second interface heartbeat; show source-event and retrieval times. Asset and entrance level only; do not use caller, staff or personal-device location. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Geocoding errors, duplicate room names, stale panel labels and an uninstrumented obstruction can make the apparent match wrong. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. emergency-service mobilising control, the university address and space-data custodian, Security Control and the on-scene site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the live incident record, current postal and geospatial address register, current emergency access plan, panel asset register and a direct radio or visual entrance confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an address, panel or entrance mapping has not been revalidated after a building, access or naming change, or a live entrance status is more than 5 minutes old. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + recommendation (options only) - identity and entrance reconciliation with ranked alternatives. Decision interval. at call receipt, mobilisation, any incident-location amendment and first-arrival confirmation. Minimum evidence. one incident reference, one current site identifier and two independent sources that agree on the building or explicitly expose the disagreement. Method. normalise addresses and aliases, cross-check the alarm panel asset against the site boundary, compare the nominated entrance with the current access plan, and present the best-supported match plus unresolved alternatives. Decision discipline. Options only; command decides. Uncertainty. state which element is confirmed, reported or conflicting, and show map age and entrance-status age separately. Safe failure. return BUILDING OR ENTRANCE NOT CONFIRMED and pass the choice to mobilising control and incident command.

ANSWER BOUNDARY

Do not infer a caller position, issue turn-by-turn driving directions, guarantee that an entrance is usable, or let an alarm-panel label override emergency-service information or direct observation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Reduces wrong-site attendance and gives the first appliance a traceable entrance reference that can be checked before committing vehicles.

STAKEHOLDER QUESTION

ER-002

Which appliance gate and apron are currently reported available, and what verified width, height, turning or obstruction limits apply there?

SENSORS AND TELEMETRY

Proposed evidence. Proposed gate-position contacts, barrier-controller health, vehicle-loop occupancy, fixed clearance markers and authorised obstruction reports can describe the gate and immediate apron. Coverage. Each emergency vehicle gate, barrier, turning apron and the first section of the internal access road. Timing, aggregation and privacy. State changes as events, controller heartbeat every 60 seconds, and obstruction reports with observed and expiry times. Gate and route-segment level; no vehicle registration capture or individual movement history. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. A contact proves only the gate leaf or barrier state; it cannot prove full route width, surface loading, overhead clearance or absence of an unreported obstruction. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, Security Control, Estates transport and access management, the road or structure owner and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. direct crew confirmation, current gate dimensions and swept-path information, emergency access plan, temporary traffic-management plan, barrier fault log and authorised closure notices. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when dimensions have not been checked after works, a closure has no effective or expiry time, or live gate state is older than 3 minutes. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - gate suitability screening and conflict display. Decision interval. before appliance commitment to the site entrance and whenever a gate, route or staging condition changes. Minimum evidence. a named gate, current dimensions or restriction record, a current state or direct report, and an accountable owner for unresolved restrictions. Method. compare appliance requirements supplied by emergency-service control with gate width, height and turning constraints; overlay current closures and obstruction reports without converting the result into a driving command. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. separate measured dimensions, reported obstructions and unverified route sections; state when only the immediate gate is covered. Safe failure. show GATE SUITABILITY NOT ESTABLISHED and list the evidence gap or alternative gate for command to assess.

ANSWER BOUNDARY

Do not open or unlock a gate, authorise access, certify structural loading or clearance, identify vehicles, or describe a gate contact as proof that the full route is clear. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps mobilising control and the incident commander avoid committing large appliances to a gate that is closed, constrained or unsupported by current evidence.

STAKEHOLDER QUESTION

ER-003

Which approach-route constraints - axle load, bridge or ramp capacity, tight turns, flooding, surface condition or overhead clearance - could affect the selected appliance?

SENSORS AND TELEMETRY

Proposed evidence. Proposed water-depth gauges at known low points and road-surface temperature sensors may supplement dated surveyed clearances, widths, turning geometry and structural load limits; obstruction status remains a direct or authorised report. Coverage. The public-road interface, estate approach roads, ramps, bridges, undercrofts, tight turns and known flood-prone sections used by emergency vehicles. Timing, aggregation and privacy. Water and surface conditions at 1-minute intervals during an incident; fixed restrictions by effective period; temporary reports as events. Route-segment level only, with no tracking of emergency or private vehicles. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Point sensors do not establish the condition of an entire route, and telemetry cannot verify structural capacity, tyre grip or safe manoeuvring. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. the relevant highway or estate-road authority, Estates engineering, the structural asset owner, emergency-service control and incident command. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current route and bridge restrictions, measured clearance schedule, swept-path assessment, temporary works and traffic orders, flood response plan and direct crew observation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a restriction has passed its reassessment date, works have altered the route, or live water and surface observations are older than 5 minutes. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + comparison - constraint filtering across candidate approach routes. Decision interval. during pre-arrival planning, at severe-weather updates and before rerouting a specialist or heavy appliance. Minimum evidence. one current candidate route, verified fixed constraints for each critical segment and either recent direct or telemetry evidence for temporary conditions. Method. screen route segments against the supplied appliance envelope, flag any unknown structural or geometric constraint, and show the last confirmed condition rather than infer continuity between sensors. Decision discipline. Retain inputs, units and formula; keep conflicts visible. Uncertainty. identify unsurveyed segments, point-sensor coverage and any restriction whose applicability to the arriving appliance is unclear. Safe failure. return APPROACH CONSTRAINTS INCOMPLETE and refer route selection to emergency-service control and the incident commander.

ANSWER BOUNDARY

Do not certify a route as safe, calculate a manoeuvre for the driver, override highway restrictions, or infer load capacity from the absence of a sensor alarm. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports an early decision about whether the normal approach is credible or whether a different gate, vehicle position or specialist access check is needed.

STAKEHOLDER QUESTION

ER-004

At the agreed secure entrance, which authorised key, fob or local access resource is confirmed available, and through which custodian?

SENSORS AND TELEMETRY

Proposed evidence. Proposed key-cabinet and premises-information-box open/closed contacts, tamper alarms, electronic-lock health and role-coded access acknowledgements can report availability without exposing credentials. Coverage. Emergency gates, secure external doors, fire-service access points, key safes and controlled key cabinets relevant to response. Timing, aggregation and privacy. Event-driven for access and tamper state, with a 60-second device heartbeat and a current shift-level custodian acknowledgement. Asset and role level; never display PINs, cryptographic secrets, personal access histories or named key-holder movements. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. A cabinet or lock signal cannot prove that the correct key is present, works, has not been duplicated or will open a mechanically damaged door. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, Security Control, the responsible site representative, Estates access-control owner and the relevant fire-safety responsible person. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current emergency access procedure, key and fob custody register, premises-information-box inventory, lock maintenance record and direct confirmation from the authorised custodian. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the custodian has not acknowledged the current shift, a key inventory predates a lock change, or an electronic state is more than 3 minutes old. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - access-resource confirmation with role and asset matching. Decision interval. during mobilisation, at site arrival and whenever the nominated access point changes. Minimum evidence. a named entrance, a current authorised custodian or emergency procedure, an asset-level access record and a direct availability confirmation where practicable. Method. match the entrance to the authorised key, fob or override route; show custodian role and contact channel, not credentials; keep unverified physical availability separate from electronic status. Decision discipline. Separate observation from interpretation. Uncertainty. state whether access is confirmed, electronically indicated, reported or unknown, and identify any missing key-content check. Safe failure. show ACCESS RESOURCE NOT CONFIRMED and direct the issue to the site liaison and incident command.

ANSWER BOUNDARY

Do not disclose access codes or credential data, remotely unlock or override a system, authorise entry, identify individual access-card users or promise that a key will work. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps the first crews reach the agreed secure access point and obtain the correct access resource without unsafe credential disclosure or delay.

STAKEHOLDER QUESTION

ER-005

Which pre-planned initial rendezvous point can arriving resources report to without conflicting with appliance access, evacuation or casualty routes?

SENSORS AND TELEMETRY

Proposed evidence. Proposed external wind and weather sensors, fixed smoke and visibility instruments, anonymous area occupancy counts, gate and road-segment state, and authorised obstruction reports can describe candidate rendezvous areas. Coverage. Pre-defined external rendezvous and command-post candidates, adjoining appliance routes, pedestrian exits and casualty handover paths. Timing, aggregation and privacy. Environmental readings every 30 seconds, area counts every minute, and access or obstruction changes as events. Zone-level counts and conditions only; no facial recognition, personal-device tracking or identification of evacuees or responders. Quality and provenance. Static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Fixed sensors may be upwind of the actual plume, crowd counters can undercount, and no telemetry can establish tactical suitability or future smoke movement. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. Incident command, police and ambulance commanders as applicable, Security Control, Estates access management and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. The incident command decision, current emergency plan, traffic and event controls, weather observations, live site reports and direct inspection of the candidate location. Times, permission and provenance. Show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. Treat the answer as stale when environmental or occupancy evidence is older than 2 minutes, a route state is older than 5 minutes, or the emergency plan predates an access change. Conflicts. Show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + observation + comparison + recommendation (options only) - pre-arrival rendezvous screening. Decision interval. At mobilisation, when access or evacuation routes change, and when command confirms a different reporting point. Minimum evidence. At least one named pre-planned point, current route and obstruction evidence, separation from evacuation and casualty paths, and an incident-command constraint set. Method. Compare candidates against approach, turning, route conflict, space, communications and environmental observations; do not calculate a tactical winner. Decision discipline. Present evidence and gaps as options for command; never issue a location instruction. Uncertainty. Show age, sensor distance, unobserved boundaries and conflicts with direct reports. Safe failure. Display INITIAL RENDEZVOUS NOT VERIFIED and refer the location decision to mobilising control and incident command.

ANSWER BOUNDARY

Do not select or authorise the rendezvous or command post, predict plume movement, identify people in an area or treat low fixed-point readings as proof of safety. Return unavailable or conflicting when evidence is insufficient.

WHY THIS IS USEFUL

Allows mobilising control to give arriving resources one checkable reporting point before a command post is fully established, while reducing avoidable route conflicts.

STAKEHOLDER QUESTION

ER-006

Is the premises information box physically accessible, and which required information and key groups were last confirmed present and current?

SENSORS AND TELEMETRY

Proposed evidence. A cabinet open/closed contact, tamper alarm and local temperature or moisture alarm may show whether the premises information box has been opened or exposed to damaging conditions. Content status must come from a signed physical inventory; do not infer document or key presence from weight, radio tags or a closed-door signal. Coverage. The premises information box, any controlled backup document cabinet and their mapped external access point. Timing, aggregation and privacy. State changes as events with a five-minute health heartbeat; reconcile the signed inventory after every authorised opening and scheduled physical check. Show only item groups, document versions and expiry or reassessment dates to authorised roles; never expose access codes, personal evacuation details or security-sensitive drawings outside need-to-know permissions. Quality and provenance. Function-test contacts and tamper alarms; retain cabinet ID, mapping version, last physical checker, signed inventory version, source-event time, retrieval time, clock offset, uptime and gaps. Limits. A healthy cabinet signal cannot prove contents are present, legible, current, complete or that any key will work. Deployment status. Proposed; return UNAVAILABLE unless the cabinet, mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. The fire-safety responsible person, Security Control, Estates document control, the information asset owner and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. The current premises-information-box inventory, signed inspection record, controlled emergency plans, key schedule, hazardous-material summary and direct box inspection. Times, permission and provenance. Show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. Treat the answer as stale when an item has no version or reassessment date, the inventory predates a material building or system change, or the box has been opened without subsequent reconciliation. Conflicts. Show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + observation + comparison - box access and signed inventory reconciliation. Decision interval. During mobilisation where available, at first site handover, after any authorised opening and at handback. Minimum evidence. A verified box location, current controlled inventory, last signed physical check or opening record, and a named custodian. Method. Compare required item groups and versions with the signed inventory and direct box inspection; show missing, expired, damaged and unverified groups separately. Decision discipline. Do not infer content from cabinet telemetry. Uncertainty. Distinguish physical access, signed inventory, document currency and key functionality. Safe failure. Display BOX OR CONTENT NOT VERIFIED and identify the site liaison or information owner needed next.

ANSWER BOUNDARY

Do not unlock the box, disclose restricted plans, personal evacuation information or credential data to unauthorised users, certify its contents or infer current documents from a cabinet signal. Return unavailable when physical evidence is insufficient.

WHY THIS IS USEFUL

Tells first crews whether the emergency information pack has a current signed inventory and which documents or keys still require direct confirmation.

STAKEHOLDER QUESTION

ER-007

What do verified fixed-point wind, smoke and visibility observations show on the practical approach sides, and where is there no coverage?

SENSORS AND TELEMETRY

Proposed evidence. Proposed weather stations, external particulate and carbon-monoxide sensors, temperature probes and optical visibility sensors can report conditions at fixed approach-side points. Coverage. Each principal external facade, emergency vehicle approach, main pedestrian access and nominated rendezvous area where installation is physically justified. Timing, aggregation and privacy. Wind at 10-second samples with 1-minute summaries; smoke and visibility at 30-second samples; show last calibration and heartbeat. Fixed-location environmental readings only; no camera-based person analysis or personal exposure estimation. Quality and provenance. record position, range, accuracy, calibration/duel date and drift checks. Align source clocks to the incident timeframe; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Point instruments may be shielded by the building, can saturate or lag, and cannot represent conditions around a corner, inside a plume or at responder breathing height. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, emergency-service monitoring teams, Estates environmental systems owner and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. direct crew observations and calibrated responder instruments first, then current weather-service observations, commissioned site telemetry, incident reports and the emergency access plan. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when wind or smoke data is older than 2 minutes, calibration is overdue, the device is faulted, or local observations conflict with the fixed sensor. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + calculation - time-aligned comparison of approach-side environmental observations. Decision interval. during mobilisation, immediately before committing crews to an approach side and after any visible smoke or wind change. Minimum evidence. one current direct or calibrated responder observation and at least one time-stamped fixed or external weather source for context. Method. display each sensor by physical location, height and age; compare trend and agreement without extrapolating a plume path; prioritise direct incident observations. Decision discipline. Retain inputs, units and formula. Uncertainty. show spatial separation, sensor health, calibration age and whether readings are below detection, within range or saturated. Safe failure. return APPROACH CONDITIONS NOT ESTABLISHED and rely on direct observation and incident-command assessment. Forecast boundary: do not turn a trend or aligned sequence into a forecast without a verified model, horizon and error bound.

ANSWER BOUNDARY

Do not predict plume travel, recommend an approach, estimate personal exposure, certify air as safe or let fixed sensors override calibrated responder instruments. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command compare approach sides and position initial resources while understanding exactly where environmental evidence is absent or weak.

STAKEHOLDER QUESTION

ER-008

Which boundary-to-door route is currently evidenced for ambulance crews, stretcher teams and responders on foot, including step-free constraints?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door-position contacts, gate state, ramp and threshold flood sensors, lift and platform-lift status, emergency-lighting health and route obstruction reports can describe nominated access paths. Coverage. Accessible pedestrian entrances, ambulance handover points, ramps, wide doors, principal corridors and lifts that form pre-planned responder routes. Timing, aggregation and privacy. Door and lift changes as events, environmental state every minute, and manual obstruction reports with observed and expiry times. Route-segment and asset level; no tracking of patients, staff, visitors or responders. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timeframe and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. A route can contain uninstrumented debris, smoke, gradients, narrow turns or damaged surfaces; a door-open signal does not prove clear passage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. ambulance incident command, the fire incident commander, Estates access and lift owners, Security Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. direct route inspection, current accessible access plan, emergency plan, lift and door maintenance status, temporary closure notices and authorised site reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a planned route has not been revalidated after works, any critical asset state is older than 3 minutes, or a manual route report is older than 10 minutes during active change. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - route-segment availability screening for foot and stretcher access. Decision interval. before selecting a responder foot route, before casualty movement and whenever smoke, water, lift or door conditions change. Minimum evidence. a named origin and destination, current state for every critical door or lift, one recent direct or accountable route report and known dimensional constraints. Method. compose only pre-defined route segments, flag unsupported segments and conflicting states, and present alternatives for command assessment rather than optimise a route. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state where coverage stops, whether dimensions are measured or assumed, and which conditions are direct, reported or sensor-indicated. Safe failure. show ROUTE PASSABILITY NOT CONFIRMED and require direct assessment by the relevant commander.

ANSWER BOUNDARY

Do not direct casualty movement, certify an accessible route, infer patient needs, operate lifts or doors, or represent telemetry as proof that a route is clear. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports early coordination between fire and ambulance teams by identifying which planned access route has current evidence and where a physical check is still required.

STAKEHOLDER QUESTION

ER-009

Can the required specialist appliance reach and leave the service side without blocking fire access, evacuation or casualty movement?

SENSORS AND TELEMETRY

Proposed evidence. Proposed yard bay-occupancy sensors, gate contacts, loop detectors, fixed clearance markers, anonymous area counts and authorised loading or obstruction reports can describe the service-side approach. Coverage. The loading yard, service road, delivery gates, turning heads, fire access lanes, evacuation exits and casualty transfer interfaces. Timing, aggregation and privacy. Occupancy and gate state every 30 seconds or on change; reports and bookings by effective and expiry time. Bay and route-segment level; no number-plate recognition or persistent vehicle tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Sensors cannot prove pavement load capacity, safe reversing, full turning clearance or that a temporarily stationary vehicle will remain in place. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, emergency-service logistics, Estates transport and loading-bay management, Security Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current service-yard plan, swept-path and clearance records, fire access requirements, delivery and event controls, live site reports and direct vehicle marshal observation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a yard plan predates physical works, a booking or closure has no current effective time, or bay and gate state is older than 3 minutes. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - access-conflict screening for specialist vehicle deployment. Decision interval. before dispatching or committing a specialist vehicle, and whenever staging, evacuation or casualty routes are reconfigured. Minimum evidence. vehicle requirements supplied by the service, a current yard layout, status of critical gates and bays, and command-defined routes that must remain unobstructed. Method. compare vehicle envelope and support needs with fixed constraints and current occupancy; flag conflicts with protected fire, evacuation and casualty routes; leave final positioning to command and marshals. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. identify unmeasured turning areas, unverified surface capacity and occupancy observations without persistence guarantees. Safe failure. return SERVICE-SIDE ACCESS NOT ESTABLISHED and identify the missing check or alternate holding area.

ANSWER BOUNDARY

Do not direct vehicle manoeuvres, reserve or clear bays, certify loading capacity, identify drivers or vehicles, or authorise obstruction of a protected access route. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps logistics leads to decide whether specialist resources should be held, redirected or physically marshalled before they enter a constrained service area.

STAKEHOLDER QUESTION

ER-010

Which temporary works, events, parked plant, barriers or closures have changed emergency access since the latest controlled access plan?

SENSORS AND TELEMETRY

Proposed evidence. Proposed gate or door contacts, temporary barrier beacons and georeferenced authorised inspection reports may indicate selected changes; temporary-works, event and closure records remain the principal evidence. Coverage. External approaches, emergency gates, fire access lanes, loading and event areas, principal entrances and any route affected by a live works permit. Timing, aggregation and privacy. Physical state on change with a 5-minute heartbeat; permits, events and closures by effective, reassessment and expiry times. Asset and route-zone level; no worker or attendee identification. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Unregistered works, moved barriers and incomplete permit closure can leave material changes invisible; telemetry cannot establish the legal or engineering adequacy of temporary arrangements. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. Estates projects and maintenance, event management, Security Control, the principal contractor, fire-safety management and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current permit-to-work and change-control registers, temporary traffic and fire-safety plans, event operations plan, closure notices, contractor handover and direct site inspection. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a permit or closure lacks an effective or expiry time, the site has changed since the last inspection, or a temporary-state report is older than 10 minutes. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + diagnosis - change detection and impact mapping against emergency access requirements. Decision interval. at mobilisation, first site handover, each shift change and whenever a work, event or closure status changes. Minimum evidence. a current baseline access plan, all active authorised changes for the incident period and at least one accountable or direct confirmation of critical temporary arrangements. Method. compare current authorised changes with the baseline, map affected gates and route segments, and display unverified or expired controls as conflicts rather than assume removal. Decision discipline. Root cause needs competent confirmation; keep conflicts visible. Uncertainty. show changes known only from records, changes known only from field reports, and areas not recently inspected. Safe failure. state CURRENT ACCESS CHANGES INCOMPLETE and require direct confirmation before relying on the baseline plan.

ANSWER BOUNDARY

Do not approve temporary works, certify fire access, infer that an expired permit means the obstruction has gone, or let the May 2021 plan establish current access. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents responders relying on an apparently familiar access plan when construction, an event or a temporary closure has materially changed the site.

STAKEHOLDER QUESTION

ER-011

Where is the incident reported or detected inside the building, and which evidence supports or conflicts with that location?

SENSORS AND TELEMETRY

Proposed evidence. Proposed fire-alarm device events, sprinkler flow zones, fixed gas or water alarms, room-level environmental sensors and manually entered responder or site reports can contribute location evidence. Coverage. All commissioned alarm zones and mapped devices, plus selected plant, laboratory and service areas with relevant fixed detection. Timing, aggregation and privacy. Event-driven with original panel time, gateway receipt time and responder-service retrieval time preserved; manual reports keep observed time. Device, room or zone level as authorised; no person or personal-device location. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Alarm zones can span several rooms, devices can be wrongly labelled or triggered by spread rather than origin, and fixed sensors cannot locate an incident between coverage points. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service control, followed by the fire-alarm responsible person, relevant system owner, Security Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current incident log, direct crew reports, alarm panel and suppression logs, current alarm-zone schedule, room and floor register, and verified system drawings. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a device-to-room mapping predates alterations, a manual report has been superseded, or a live system event is not accompanied by current panel health and clock status. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + estimate - location evidence fusion with explicit source weighting. Decision interval. at initial handover, after each new alarm or crew report, and before changing sectors or search priorities. Minimum evidence. one emergency-service or direct report plus one independently mapped building-system or authoritative location source, or an explicit statement that only one source exists. Method. map each event and report to a common floor-zone-room vocabulary, order by time and precedence, and show the supported area, competing areas and reason for disagreement. Decision discipline. Label assumptions and range. Uncertainty. use bounded areas rather than a false point location; show zone size, mapping age, clock quality and whether evidence may indicate spread rather than origin. Safe failure. return INCIDENT LOCATION UNCONFIRMED and preserve all reported alternatives for incident command.

ANSWER BOUNDARY

Do not claim the seat, cause or origin of an incident, locate a person, direct tactical entry, or let a derived confidence score override crew reports or command. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives incident command a common, time-stamped picture of where evidence points and where the location remains too broad or conflicted for reliance.

STAKEHOLDER QUESTION

ER-012

From the selected entry, which pre-planned route reaches the reported incident area, and which segments still require direct confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door and shutter contacts, smoke and heat sensors, water-depth or leak sensors, emergency-lighting health, lift status and authorised obstruction reports can describe critical route segments. Coverage. Only named pre-planned emergency routes linking responder entrances to stairs, fire-fighting shafts, lobbies and incident zones. Timing, aggregation and privacy. Asset changes as events, environmental data every 30 seconds to 1 minute, and manual route observations with time and observer role. Route-segment and asset level; no responder tracking or inferred crew position. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Coverage gaps, smoke stratification, uninstrumented debris and rapidly changing fire conditions mean a route can become unusable between updates. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, direct crews, the fire-safety responsible person, Estates building and lift owners and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current emergency and fire-fighting access plans, verified as-built route information, system status, temporary closure notices and direct route reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a route plan predates layout change, a critical status is older than 2 minutes during active conditions, or a manual observation is more than 5 minutes old without reconfirmation. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - segment-by-segment evidence display for named route options. Decision interval. before committing to a planned route, after any new smoke, door, lift or obstruction event, and at sector handover. Minimum evidence. a confirmed origin and target zone, a current route definition, evidence for every critical barrier or vertical element, and direct observation for any unsupported segment. Method. assemble only controlled route segments, attach current evidence and age to each segment, expose gaps and alternatives, and never convert the result into a tactical routing instruction. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state spatial coverage, sensor height, unsupported segments, and the possibility that conditions changed after observation. Safe failure. show ROUTE CONDITIONS INCOMPLETE and require direct crew assessment under incident command.

ANSWER BOUNDARY

Do not choose or command an entry route, predict fire or smoke development, certify a route clear, operate doors or lifts, or infer crew location from route queries. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps crews and command to see where a planned route is supported by current evidence and where reconnaissance or a different access check is required.

STAKEHOLDER QUESTION

ER-013

How do the alarm zone, device address, room number, floor name, stair label and emergency-service sector name map to one another?

SENSORS AND TELEMETRY

Proposed evidence. Proposed panel and gateway interfaces can expose the device, loop and zone identifiers exactly as recorded; fixed QR or NFC location labels can support anonymous visual confirmation at key junctions. Coverage. All alarm zones, rooms, levels, stairs, fire-fighting shafts, plant areas and external sector reference points used in emergency information. Timing, aggregation and privacy. Mappings are version-controlled; live events retain original identifiers and timestamps; label checks occur after changes and scheduled inspections. Location-reference level only; anonymous tag reads are not retained as movement history. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. Legacy aliases, reused room numbers and undocumented local labels can create false equivalence; a tag may be missing, damaged or placed on the wrong side of a boundary. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational sector names, Estates space-data management, the fire-alarm system owner, fire-safety management and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current room and floor register, alarm-zone schedule, stair and shaft schedule, emergency plan, incident sector plan and direct label confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a mapping has not been revalidated after room renumbering, panel reprogramming or compartment change, or two current systems use unresolved names for the same place. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison - crosswalk validation between operational and building naming systems. Decision interval. at initial handover, whenever a new zone or sector is declared, and when crews report a naming mismatch. Minimum evidence. the exact source identifier, one current authoritative location register, and a command-approved sector name or explicit absence of one. Method. preserve original labels, map them through a versioned crosswalk, display one-to-many relationships and unresolved aliases, and never silently replace source terminology. Decision discipline. Keep conflicts visible. Uncertainty. show mapping confidence, version, ambiguous boundaries and any label verified only from historical material. Safe failure. return LOCATION NAMES NOT RECONCILED and display all source labels side by side.

ANSWER BOUNDARY

Do not rename sectors for command, infer equivalence from similar wording, track tag users, or let the May 2021 floor plan establish current room or floor naming. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents different agencies and site teams talking about the same place with incompatible labels during tasking, search and handover.

STAKEHOLDER QUESTION

ER-014

Which independent fallback routes remain available if the primary corridor, stair or lobby is unusable, and which hidden dependencies do they share?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door, shutter and damper contacts, smoke and temperature sensors, stair and lobby pressure sensors, water detection, emergency-lighting status and authorised blockage reports can inform alternatives. Coverage. Pre-defined alternate routes between emergency entrances, stairs, fire-fighting lobbies and the affected floor or zone. Timing, aggregation and privacy. Environmental data every 30 seconds, asset state on change and manual observations with observed and expiry times. Route-segment level with no responder, occupant or device tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Not all route segments will be instrumented, conditions can change faster than reporting, and a nominal alternative may share the same hidden void, smoke route or access dependency. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then fire-safety management, Estates building and lift owners and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current emergency access and fire strategy information, route and compartment drawings, live life-safety system status, closures and direct reconnaissance reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an alternative route has not been checked after alterations, any critical environmental state is older than 2 minutes, or a field report has been superseded. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + recommendation (options only) - dependency-aware comparison of pre-defined alternative routes. Decision interval. when the primary route is reported blocked, after material system or smoke changes and at each operational reassessment. Minimum evidence. two named route options, a current target zone, evidence for shared dependencies and at least one recent direct report for the chosen access area. Method. compare routes by independent segments and shared barriers, expose common-mode failures, and show evidence age and gaps without ranking tactical preference. Decision discipline. Options only; command decides. Uncertainty. state which segments lack sensors or recent observation and whether routes converge through the same stair, lobby, riser or external access point. Safe failure. show NO ALTERNATIVE ROUTE VERIFIED and defer route selection to incident command and reconnaissance.

ANSWER BOUNDARY

Do not designate a safe route, calculate evacuation or entry instructions, predict tenability, operate life-safety systems or infer that absence of an alarm means a segment is usable. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command avoid treating a nominal alternative as independent when it shares the same blocked or uncertain dependency as the primary route.

STAKEHOLDER QUESTION

ER-015

Which fire-fighting stair, shaft and lobby serve the affected level, and what is known about each dependency now?

SENSORS AND TELEMETRY

Proposed evidence. Proposed stair and lobby smoke, temperature and pressure sensors, fire-door contacts, fire-fighting lift state, lighting health and shaft communication-system status can provide asset evidence. Coverage. Each designated fire-fighting shaft, stair enclosure, fire-fighting lobby, associated lift and access door serving Abacws. Timing, aggregation and privacy. Environmental data every 30 seconds, doors and lift states as events, and system health heartbeat every minute. Named shaft, level and asset only; no tracking of crews or occupants using the shaft. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Fixed readings cannot establish conditions throughout the enclosure, door contacts cannot show obstruction or integrity, and lift telemetry cannot prove operational safety. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, the fire-safety responsible person, lift and smoke-control system owners, Estates engineering and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current fire strategy, verified shaft and stair schedule, lift and smoke-control status, impairment records, maintenance reports and direct lobby or stair observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the strategy or shaft mapping predates alteration, a critical status is older than 2 minutes, or any system is in fault, test or unknown health. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - service-to-level mapping and current shaft-status reconciliation. Decision interval. at initial sector planning, before reliance on a shaft and after any door, lift, smoke-control or power change. Minimum evidence. a verified level-to-shaft mapping, current status of critical doors, lift and smoke-control dependencies, and direct confirmation at the access point. Method. present design intent separately from current indication and crew observation; flag each dependency as confirmed, reported, faulted or unavailable. Decision discipline. Root cause needs competent confirmation. Uncertainty. show sensor locations, unobserved sections and any conflict between design records and current physical configuration. Safe failure. state FIRE-FIGHTING SHAFT STATUS NOT ESTABLISHED and require direct assessment.

ANSWER BOUNDARY

Do not authorise use of a shaft or lift, certify protection or tenability, operate recall or smoke control, or imply that design intent describes current conditions. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives sector commanders a concise check of which vertical access is intended to serve the level and which critical dependencies still need direct verification.

STAKEHOLDER QUESTION

ER-016

Which planned casualty-extraction route links the reported location to an ambulance handover point, and where are the limiting widths, gradients, doors or lift dependencies?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door and gate contacts, corridor and stair smoke or water sensors, lift status, route lighting health, anonymous congestion counts and authorised obstruction reports can describe pre-planned casualty routes. Coverage. Named routes between internal zones, fire-fighting or evacuation lifts where applicable, accessible exits and external ambulance handover points. Timing, aggregation and privacy. Asset state on change, environmental and congestion summaries every minute, and manual reports with observed time. Route-zone level; no identification, patient tracking, health monitoring or individual movement history. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Telemetry cannot determine patient condition, stretcher handling space, required clinical resources or whether a route remains tenable during movement. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. ambulance and fire incident command, direct crews, Estates access and lift owners, Security Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. command decisions, current casualty movement and accessible access plans, verified dimensions, lift and door status, live route reports and external handover arrangements. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when route dimensions or plans predate works, any critical state is older than 2 minutes, or the ambulance handover point has changed without record update. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - constraint mapping for nominated casualty movement options. Decision interval. when a casualty location is reported, before movement begins and after any route, lift, smoke or external handover change. Minimum evidence. a bounded casualty zone, an ambulance-approved handover point, current status of every critical segment and a direct route assessment where sensing is incomplete. Method. display route dimensions, barriers, vertical dependencies, current hazards and external conflicts; do not optimise or instruct movement; show alternatives only as evidence maps. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. identify unsupported segments, rapidly changing conditions and any dimensional information taken from unverified or historical drawings. Safe failure. return CASUALTY ROUTE NOT CONFIRMED and defer movement planning to ambulance and fire command.

ANSWER BOUNDARY

Do not infer patient condition or survival, direct casualty movement, certify a route, allocate clinical priority, operate lifts or disclose identifiable casualty information. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports joint fire-ambulance planning by exposing practical route constraints before a casualty movement is attempted.

STAKEHOLDER QUESTION

ER-017

Which stable, visible landmarks should be used to name sectors, cordons and reporting points consistently?

SENSORS AND TELEMETRY

Proposed evidence. Fixed reflective, illuminated or tactile location markers and passive QR or NFC labels may provide stable names for selected entrances, stairs, external corners and plant access points. No automatic reader network or responder-device tracking is required. Coverage. Only landmarks approved for the emergency plan and physically visible from the intended reporting position; all other areas remain unmarked. Timing, aggregation and privacy. Static lookup, with condition recorded after each authorised inspection and after works or damage; a crew-initiated scan returns the marker ID without retaining device identifiers or movement history. Quality and provenance. Verify position by dated survey, check marker legibility and illumination, record map version, inspection time, clock basis and any missing or moved marker. Limits. GNSS may be poor near buildings, a marker can be obscured or displaced, and a landmark does not establish conditions beyond the point seen. Deployment status. Proposed; return UNAVAILABLE unless placement, mapping, inspection, permission and current condition are verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for sector definitions, Estates surveying and space data, fire-safety management, Security Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current surveyed base map, emergency plan, controlled landmark register, sector plan created for the incident and direct visual confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a landmark has not been inspected after building works, its label differs across current records, or its physical visibility is unconfirmed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - landmark selection support and naming consistency check. Decision interval. during initial sectorisation, after cordon changes and whenever a location description is misunderstood. Minimum evidence. a current surveyed location, a visible or directly confirmed marker and a command-approved operational name. Method. present durable landmarks with coordinates, photographs or descriptions only to authorised users; preserve command naming and expose conflicting site labels. Decision discipline. Separate observation from interpretation. Uncertainty. state survey date, physical confirmation date, indoor positioning limits and whether a landmark is visible from the intended approach. Safe failure. show LANDMARK NOT VERIFIED and use plain direct descriptions agreed by command.

ANSWER BOUNDARY

Do not track responders, store device identifiers, treat a landmark as route-clearance evidence or allow an old label to override direct observation and incident command.

WHY THIS IS USEFUL

Gives all agencies common, physically checkable reference names for sectors and reporting points, reducing radio ambiguity without tracking responders.

STAKEHOLDER QUESTION

ER-018

How can a crew confirm a junction, stair or plant-area location using fixed labels without the service identifying or tracking the crew?

SENSORS AND TELEMETRY

Proposed evidence. Proposed static visual codes, tactile labels and privacy-preserving NFC tags can return the fixed location identifier and nearby asset names; the service need not collect device identifiers. Coverage. Selected complex junctions, stair landings, fire-fighting lobbies, plant entrances, basement routes and roof access points. Timing, aggregation and privacy. Location content is version-controlled and checked after changes; each query returns the content version and retrieval time. No location-query logs beyond short-lived anonymous service-health counts; no device ID, user account or movement sequence. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. A damaged, duplicated or misplaced label can mislead, and the displayed location does not establish route safety or the condition beyond the immediate marker. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crew observation, then Estates space-data and signage owners, fire-safety management and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current location-marker register, room and stair schedule, controlled emergency map, inspection history and direct visual comparison with surroundings. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a marker has not been revalidated after nearby works, its content version is not current, or the physical label and authoritative register disagree. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - anonymous fixed-point location lookup with integrity validation. Decision interval. on demand when a crew needs a location reference and during scheduled marker inspection. Minimum evidence. a physically observed marker, a matching current register entry and one surrounding fixed feature that corroborates the location. Method. return the fixed location, floor, stair or plant identifier and nearby landmarks; verify tag signature or checksum; do not build a session trail. Decision discipline. Separate observation from interpretation. Uncertainty. show marker inspection date, mapping version and any mismatch or inability to verify the physical placement. Safe failure. return LOCATION MARKER UNVERIFIED and require confirmation from direct features, plans and command.

ANSWER BOUNDARY

Do not identify the user, store device identifiers, reconstruct movements, infer crew position between queries, or represent the marker as proof of safe access. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps crews resolve confusing internal locations while preserving operational privacy and avoiding a responder-tracking system.

STAKEHOLDER QUESTION

ER-019

Where are the verified access points and competent owners for the roof, basement, plant rooms and service voids relevant to this incident?

SENSORS AND TELEMETRY

Proposed evidence. Proposed access-door contacts, key-cabinet status, local smoke or gas alarms, lighting health and environmental condition sensors can describe selected high-risk access points. Coverage. Roof hatches and stairs, basement entrances, principal plant-room doors, riser cupboards and service-void access panels included in the emergency information set. Timing, aggregation and privacy. Access and alarm changes as events, health heartbeat every minute and controlled access-point inspection after works. Asset and access-point level; no access-card user identity or movement history. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. A door contact cannot show what lies beyond, hidden void access may be unrecorded, and records may omit contractor-created openings or locked internal panels. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, Estates engineering and space data, relevant electrical or mechanical responsible persons, Security Control and fire-safety management. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current verified as-built and plant access records, key schedule, hazard register, permit and change-control records, access-door status and direct inspection. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an access location has not been checked after works, keys or locks have changed, or current hazard and access records disagree. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - incident-relevance filtering of controlled specialist access points. Decision interval. during sector planning, before specialist entry and whenever the incident expands into a new level or service zone. Minimum evidence. a current bounded incident area, a verified access-point register, relevant hazard ownership and direct confirmation of the selected access point. Method. filter access points by spatial and service connection to the incident, present route and key dependencies, and separate recorded location from current physical accessibility. Decision discipline. Separate observation from interpretation. Uncertainty. show survey and inspection age, hidden or unsurveyed voids and any access point known only from historical drawings. Safe failure. state SPECIALIST ACCESS NOT VERIFIED and require competent site and crew confirmation.

ANSWER BOUNDARY

Do not authorise entry, unlock doors, certify confined-space or electrical safety, infer hidden void extent, or disclose sensitive access information outside approved roles. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps specialist teams to identify the correct access point and competent owner before committing time to a wrong, locked or hazardous plant route.

STAKEHOLDER QUESTION

ER-020

Which walls, doors, room subdivisions or circulation routes are known or suspected to differ from the current controlled emergency map?

SENSORS AND TELEMETRY

Proposed evidence. Controlled post-works laser, photographic or measured surveys may flag changes to doors, partitions and circulation geometry; door-state data can only indicate selected current openings. Controlled drawings, completion records and direct observation remain primary. Coverage. Surveyed emergency routes, compartment boundaries and access points included in an authorised change package; concealed construction and unsurveyed rooms remain gaps. Timing, aggregation and privacy. Survey after material works and before controlled map publication; event-driven door states where commissioned. Exclude people and research content from imagery, and retain only authorised building geometry. Quality and provenance. Record survey method, control points, positional accuracy, date, responsible surveyor, drawing version, clock offset for live states, uptime, gaps and source-to-map transformations. Limits. Surveys may miss concealed construction, temporary partitions or locked spaces; geometry alone cannot establish fire performance, structural capacity or lawful room use. Deployment status. Proposed; return UNAVAILABLE unless survey scope, mapping, permission and current source health are verified.

AUTHORITATIVE SUPPORTING DATA

Owners. Estates project and space-data owners, fire-safety management, the principal contractor, building control or competent designers where applicable, and incident command. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current change-control and completion records, verified as-built drawings, fire strategy updates, room register, temporary works plans and direct crew observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when drawings predate a completed change, works are open without an updated emergency plan, or observed layout conflicts with the controlled record. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + diagnosis - map-to-current-state discrepancy screening. Decision interval. at incident handover, before relying on a detailed route or compartment boundary, and after crews report a mismatch. Minimum evidence. the controlled current map version, all open or recently completed change records for the area and at least one direct or verified survey observation. Method. compare expected geometry and assets with authorised change records and observations, highlight the affected area, and never infer fire resistance or current room use from geometry alone. Decision discipline. Root cause needs competent confirmation; keep conflicts visible. Uncertainty. show survey coverage, inaccessible areas, unapproved-change risk and the status of each discrepancy. Safe failure. mark MAP NOT RELIABLE FOR THIS AREA and use direct reconnaissance and current competent advice.

ANSWER BOUNDARY

Do not certify as-built accuracy, infer room use or fire performance, inspect research content, or let the May 2021 floor plan establish current walls, doors or routes. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Warns crews and commanders before they rely on a route or compartment line that may have been altered since the emergency information was last verified.

STAKEHOLDER QUESTION

ER-021

Which hazardous substances, dangerous processes or stored materials are currently registered in or beside the affected area, and what has been directly confirmed?

SENSORS AND TELEMETRY

Proposed evidence. Proposed fixed gas, vapour, oxygen-deficiency, temperature, leak and cabinet-condition sensors can indicate selected hazards where a competent risk assessment justifies them. Coverage. Only mapped laboratories, stores, workshops, plant areas and transfer points with registered substances or processes; coverage must name the monitored hazard and detection boundary. Timing, aggregation and privacy. Alarm events immediately, trend data every 30 seconds to 1 minute, and device health and calibration status with every retrieval. Room, store or process-zone level; no research-content inspection, experiment monitoring, staff identification or inference from procurement alone. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Sensors detect only specified substances within range and can cross-react, saturate or miss sealed containers; a register may not reflect recent movement or an unreported process. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and specialist hazardous-material advisers, then the laboratory or process responsible person, Health and Safety, Estates engineering and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current hazardous-substance and process register, COSHH or equivalent risk information, safety data sheets, local emergency arrangements, inventory change records and direct competent confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a register lacks a recent owner reassessment, material has moved or a process changed, an SDS version is superseded, or sensing is uncalibrated, faulted or outside its stated coverage. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - hazard-register filtering and sensor-to-registered-hazard reconciliation. Decision interval. at initial hazard handover, before entry or escalation into the area, and whenever new containers, labels, readings or process information are reported. Minimum evidence. a bounded incident area, a current accountable register owner, substance or process identifiers with quantities or ranges where authorised, and direct or instrument evidence for any claimed release. Method. filter registered hazards by location and plausible connection, distinguish presence from release, attach SDS and emergency-reference links for authorised roles, and display discrepancies without guessing content. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state inventory reassessment age, location precision, sensor specificity and whether the hazard is registered, reported or detected. Safe failure. return HAZARD INFORMATION INCOMPLETE and escalate to the competent site owner and emergency-service specialist advice.

ANSWER BOUNDARY

Do not infer unregistered chemicals, disclose sensitive research details, estimate personal dose or survival, certify an atmosphere, or issue handling, entry or neutralisation commands. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives crews an accountable starting point for identifying specialist advice, protective information and exclusion needs without confusing stored presence with an active release.

STAKEHOLDER QUESTION

ER-022

Where are the relevant electricity, gas, water and ventilation isolation points, who is competent to operate them, and what state is only indicated rather than proved?

SENSORS AND TELEMETRY

Proposed evidence. Proposed switch, breaker, valve and damper auxiliary contacts, controller feedback and local indicator capture can report intended or indicated state; manual actions require role-coded confirmation. Coverage. Main and local isolation points mapped to the affected zone, including electrical distribution, gas, water, mechanical ventilation and specialist services where documented. Timing, aggregation and privacy. State changes as events, controller heartbeat every 30 to 60 seconds, and manual confirmation with observed time and responsible role. Asset state and served-zone level; no operator identity beyond the accountable role required by the authorised action record. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. Remote indication can be wrong because of failed feedback, backfeeds, bypasses, downstream storage or undocumented cross-connections; it does not prove de-energisation or safe isolation. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, the electrical authorised person, gas responsible person, mechanical and water system owners, Estates Control and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current single-line and service schematics, isolation schedules, permits and lock-off records, controller event logs, local indicator observations and competent-person test evidence. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the schematic predates alteration, a state lacks current feedback or local verification, a permit is expired or unclear, or an isolation has no identified competent owner. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - service-to-zone tracing and indicated-versus-verified state reconciliation. Decision interval. before requesting or relying on an isolation, after any authorised change and at every operational and recovery handover. Minimum evidence. a uniquely identified isolation asset, current served-area mapping, indicated state, competent owner and physical verification or test evidence where safety depends on isolation. Method. trace the service dependency from source to affected zone, display command, feedback and direct test separately, and expose possible backfeeds, bypasses and unknown downstream energy. Decision discipline. Root cause needs competent confirmation. Uncertainty. label each state as commanded, indicated, physically observed or proven by competent test; show mapping and observation age. Safe failure. state ISOLATION NOT VERIFIED and require action and confirmation through incident command and the competent system owner.

ANSWER BOUNDARY

Do not operate or isolate any system, authorise an isolation, certify de-energisation, disclose restricted control details or let telemetry replace competent testing and lock-off procedures. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command identify the correct competent owner and isolation point while preventing a remote indication being mistaken for a safe isolation.

STAKEHOLDER QUESTION

ER-023

Which photovoltaic, battery, UPS or generator systems could remain energised, hot, fuelled or gas-producing after normal supplies are lost?

SENSORS AND TELEMETRY

Proposed evidence. Proposed inverter state, string current, battery temperature and off-gas detection, UPS alarms, generator run state, fuel-leak detection and enclosure temperature can indicate selected asset conditions. Coverage. Each registered photovoltaic array, inverter, battery energy-storage system, UPS room, standby generator and associated fuel store or cable route relevant to Abacws. Timing, aggregation and privacy. Critical alarms as events, operating values every 30 seconds, and health, calibration and communications status with every answer. Asset and enclosure level; no inference about equipment not in the verified asset register. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Systems may remain energised despite shutdown commands, sensors can be lost in fire, and remote status cannot establish absence of voltage, thermal runaway, flammable gas or concealed cable damage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service electrical specialists, then the electrical authorised person, Estates energy and generator owners, maintenance contractor and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current asset register, electrical single-line diagrams, manufacturer emergency information, maintenance and fault records, fuel and battery hazard assessments, live controller status and competent test results. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an asset or cable route has changed without drawing update, controller data is older than 1 minute, maintenance is overdue, or health and direct observations conflict. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - distributed-energy hazard inventory and state correlation. Decision interval. at hazard briefing, before work near the asset or connected cabling, after any isolation attempt and throughout prolonged or recovery operations. Minimum evidence. verified asset identity and location, current connection and isolation information, live or direct condition evidence and access to the competent electrical owner. Method. trace sources and possible backfeeds, separate normal, alarm, fault and unknown states, display thermal and gas trends with sensor range, and link authorised emergency information. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unmonitored components, sensor survival, hidden DC routes, stored-energy duration and whether remote status has local confirmation. Safe failure. return DISTRIBUTED-ENERGY HAZARD NOT CHARACTERISED and maintain the asset as an unknown live hazard for command assessment.

ANSWER BOUNDARY

Do not command shutdown, certify absence of voltage or gas, predict battery failure, recommend firefighting tactics or represent inverter-off status as safe isolation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Alerts responders to persistent energy and thermal hazards that can remain after normal supplies appear to be disconnected.

STAKEHOLDER QUESTION

ER-024

Which compressed gases, cryogenics, pressure vessels or piped laboratory gases are registered near the incident, and which alarms or supply states are current?

SENSORS AND TELEMETRY

Proposed evidence. Proposed manifold pressure, cylinder-store temperature, oxygen-deficiency, flammable or toxic-gas detection, cryogenic vessel pressure and gas-panel alarms can provide selected condition evidence. Coverage. Registered cylinder stores, gas cabinets, manifold rooms, piped-gas distribution zones, cryogenic stores and pressure vessels connected to the incident area. Timing, aggregation and privacy. Alarm events immediately, process values every 30 seconds and calibration, bump-test or maintenance status with each retrieval. Asset, manifold or room level; no inference about research activity, users or cylinder contents beyond authorised registered data. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Pressure can remain after isolation, sensors detect only specified gases and locations, labels or inventories can be wrong, and a cylinder may have moved without electronic evidence. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and hazardous-material specialists, then the laboratory responsible person, gas-system responsible person, Health and Safety, Estates engineering and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current cylinder and vessel inventory, piped-gas schematics, pressure-system records, SDS and emergency sheets, maintenance and inspection records, alarm logs and competent local confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when inventory reconciliation is overdue, a cylinder or process has moved, an alarm sensor is outside calibration, or the registered gas and physical label disagree. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - gas and pressure-hazard reconciliation by location and connection. Decision interval. at initial hazard briefing, before entering or ventilating the area, after any system state change and during recovery inspection. Minimum evidence. a current inventory or process owner confirmation, location and connection map, relevant alarm or direct instrument evidence and known sensor detection limits. Method. map vessels and gas services to the affected zone, distinguish stored presence, connected supply, alarm and confirmed release, and show quantities only to authorised roles. Decision discipline. Root cause needs competent confirmation. Uncertainty. state inventory age, possible mobile cylinders, sensor specificity, residual line pressure and unmonitored spaces. Safe failure. state GAS OR PRESSURE HAZARD NOT CONFIRMED and obtain specialist assessment through incident command.

ANSWER BOUNDARY

Do not identify research work, operate valves, certify depressurisation, estimate blast or exposure outcomes, or advise entry, venting or cylinder handling. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports early specialist escalation and prevents responders treating a room as hazard-free because a fixed alarm is silent or a supply is reportedly closed.

STAKEHOLDER QUESTION

ER-025

What current asbestos, structural, facade and passive-fire information is needed before responders cut, breach, anchor or load a building element?

SENSORS AND TELEMETRY

Proposed evidence. Proposed structural movement, vibration, temperature and facade-displacement monitoring can supplement observations where installed; these sensors cannot replace surveys, drawings or competent assessment. Coverage. Known structural transfer zones, long spans, facades, roofs and areas with recorded asbestos or passive-fire features likely to be affected by intrusive action. Timing, aggregation and privacy. Structural values at 1-second to 1-minute intervals according to risk, alarms as events and controlled survey information by effective and reassessment date. Element or zone level; no occupational or person monitoring. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timeframe; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Normal readings do not prove capacity or stability, sensors may fail in heat, asbestos can be concealed, and drawings may not show later penetrations, damage or material degradation. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, structural and fire-engineering competent persons, the asbestos dutyholder, Estates asset owner and Health and Safety. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current asbestos register and survey, structural drawings and assessments, facade information, passive-fire and compartment records, alteration history, temporary works records and direct competent inspection. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a survey excludes the intended work area, records predate alteration or damage, monitoring is outside range or calibration, or no competent owner has confirmed applicability. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison - intrusive-action hazard retrieval and applicability screening. Decision interval. before any planned cutting, breaking, fixing, breaching or unusual loading, and after significant fire, impact or structural change. Minimum evidence. the exact proposed area or element, current relevant survey and drawing coverage, known limitations and a competent-person interpretation where structural or asbestos risk is material. Method. filter records spatially and by action type, show surveyed and excluded areas, separate design information from post-incident condition, and expose missing evidence prominently. Decision discipline. Keep conflicts visible. Uncertainty. state survey type and coverage, drawing status, concealed-condition risk, sensor location and whether fire exposure may invalidate previous assumptions. Safe failure. return INTRUSIVE-WORK HAZARDS NOT ESTABLISHED and require competent assessment under incident command.

ANSWER BOUNDARY

Do not authorise cutting or loading, certify absence of asbestos or structural safety, calculate collapse likelihood, or treat stable telemetry as proof that an element is safe. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Provides responders with the right records and competent contacts before an action could release asbestos, compromise structure or breach fire protection.

STAKEHOLDER QUESTION

ER-026

Which high-voltage, rooftop and major-plant hazards or controlled exclusion areas are relevant to the proposed operational area?

SENSORS AND TELEMETRY

Proposed evidence. Proposed energisation indicators, switchgear alarms, arc-fault detection, plant run state, roof wind and ice sensors, access-door contacts and local gas or heat alarms can provide condition evidence. Coverage. High-voltage rooms and routes, transformers, major mechanical plant, roof plant compounds, fall-risk edges and authorised exclusion zones relevant to the incident. Timing, aggregation and privacy. Electrical and plant alarms as events, environmental readings every minute and controlled exclusion records by effective and expiry time. Asset and controlled-zone level; no access-card identities or staff movement history. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timeframe and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Indication does not prove isolation, rooftop conditions vary sharply, exclusion markings can be damaged, and hidden cables, stored energy or rotating equipment may remain hazardous. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, electrical and mechanical authorised persons, Estates engineering, roof and structural asset owners, Health and Safety and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current high-voltage and plant schematics, authorised-person rules, roof access plan, fall-protection and exclusion records, permits, maintenance status and direct competent observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when records predate plant or cable changes, an exclusion has no current authority or expiry, weather data is older than 5 minutes, or system state lacks local confirmation. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - hazard-zone and controlled-access information retrieval. Decision interval. during sector planning, before specialist access and after plant, power or weather changes. Minimum evidence. a bounded incident area, verified hazard and exclusion mapping, current system state where relevant and an available competent system owner. Method. map registered hazards and access dependencies, show design and current states separately, and flag any overlap between planned responder activity and a controlled zone. Decision discipline. Root cause needs competent confirmation. Uncertainty. identify unverified roof areas, hidden service routes, indicator-only states and whether environmental observations represent the actual work location. Safe failure. state PLANT OR EXCLUSION INFORMATION INCOMPLETE and require direct competent assessment.

ANSWER BOUNDARY

Do not authorise entry, operate plant, certify isolation or roof safety, identify access-card users, or calculate an exclusion distance or tactical action. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command involve the correct authorised person and avoid sending crews into a specialist hazard area based only on a generic plant-room label.

STAKEHOLDER QUESTION

ER-027

Where could fire water, floodwater or a chemical release travel through drains, floors, ducts, sumps or service routes?

SENSORS AND TELEMETRY

Proposed evidence. Proposed leak cables, floor water-depth sensors, sump level and pump state, drain flow and conductivity sensors, bund alarms and selected downstream water-quality probes can indicate movement at fixed points. Coverage. Known low points, risers, floor penetrations, plant rooms, sumps, bunds, internal and external drains and sensitive downstream spaces connected to the incident area. Timing, aggregation and privacy. Water level and flow every 30 seconds to 1 minute, threshold alarms as events and drainage mapping by controlled version. Zone, sump and drain-segment level; no person monitoring. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Water can bypass sensors through cracks, voids or undocumented connections; conductivity is non-specific, blockages change pathways and models may not represent fire damage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and environmental or hazardous-material specialists, then Estates drainage and building owners, laboratory responsible persons and the environmental compliance owner. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current drainage and service drawings, bund and containment records, environmental permits, spill plans, pump and valve status, substance information and direct field observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when drainage records predate works, a blockage or pump failure is unresolved, sensor calibration is overdue, or a pathway is based only on the historical floor plan. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + estimate - source-pathway-receptor screening with live observation overlay. Decision interval. after water application, leak or spill detection, before opening drains or changing containment, and throughout incident and recovery monitoring. Minimum evidence. a bounded source area, current drainage or service connectivity, at least one downstream observation and relevant substance or water characteristics where known. Method. trace documented connections and elevation-driven pathways, attach live levels and reports, identify unmonitored branches and sensitive receptors, and avoid deterministic flow prediction. Decision discipline. Retain inputs, units and formula; label assumptions and range. Uncertainty. show model assumptions, unknown blockages, hidden penetrations, sensor spacing and whether observed water can be attributed to the incident. Safe failure. return SPREAD PATHWAYS NOT ESTABLISHED and escalate to competent environmental and building specialists.

ANSWER BOUNDARY

Do not direct containment or discharge, certify drainage isolation, predict exact contaminant travel, infer substance identity from conductivity or operate pumps and valves. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports decisions about where to inspect, protect sensitive areas and involve environmental specialists before water or contamination spreads unnoticed.

STAKEHOLDER QUESTION

ER-028

Which compartment lines are relied upon by the current fire strategy around the incident, and which doors, shutters, dampers or penetrations have known defects or uncertain status?

SENSORS AND TELEMETRY

Proposed evidence. Proposed fire-door and shutter contacts, hold-open release state, damper position feedback, differential-pressure sensing and local fault alarms can report selected barrier indications. Coverage. Mapped compartment boundaries and critical openings surrounding the reported incident zone, including doors, shutters, dampers and service penetrations where monitored. Timing, aggregation and privacy. Position and fault changes as events, pressure every 30 seconds and inspection or impairment records by effective time. Barrier and boundary-segment level; no person movement tracking through doors. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. A closed indication does not prove latching, integrity, smoke seal or fire resistance; unmonitored penetrations and damage can defeat the intended boundary. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then the fire-safety responsible person, Estates building and smoke-control owners, maintenance contractors and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire strategy and compartment drawings, door and damper schedules, inspection and impairment records, cause-and-effect information, live system state and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the strategy predates alteration, an inspection is overdue, a barrier is faulted or damaged, or telemetry and direct observation disagree. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - design-intent and current-barrier-state reconciliation. Decision interval. at sector planning, after any barrier or smoke-control event, before relying on a compartment line and during recovery inspection. Minimum evidence. a current boundary definition, identified critical openings, recent inspection or live state and direct confirmation where tactical reliance is proposed. Method. display intended boundary, each opening and its current evidence separately; flag unmonitored penetrations, impairment and conflict; never collapse the result into a certified compartment status. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state drawing and inspection age, sensor coverage, unobserved penetrations and whether position feedback is direct or inferred from control state. Safe failure. show COMPARTMENT BOUNDARY NOT VERIFIED and require direct assessment and competent fire-safety advice.

ANSWER BOUNDARY

Do not certify compartmentation, order doors or dampers, infer fire resistance from position, track door users or promise that spread will be contained. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps crews understand the intended spread-control geometry while seeing exactly which openings or records cannot be relied on without inspection.

STAKEHOLDER QUESTION

ER-029

Which shafts, ducts, risers, voids or confined service spaces could carry smoke, heat, water or contaminants beyond the visible incident area?

SENSORS AND TELEMETRY

Proposed evidence. Proposed duct smoke detectors, riser temperature and pressure sensors, void heat or leak detection, damper feedback and selected gas sensors can indicate conditions at monitored points. Coverage. Principal ventilation ducts, vertical risers, service shafts, ceiling and floor voids, cable routes and confined service spaces connected to the incident zone. Timing, aggregation and privacy. Alarm and damper events immediately, trend values every 30 seconds to 1 minute and mapping by controlled version. Service segment, shaft or void zone; no inspection of network content, research equipment use or people. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Hidden branches and undocumented penetrations are common, point sensors can miss stratified or intermittent spread, and airflow can reverse when systems or openings change. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command. Estates mechanical and building services owners, fire-safety management, relevant laboratory or process owners and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current MEP and fire-strategy drawings, riser and duct schedules, ventilation cause-and-effect, confined-space register, alteration records, live alarms and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when drawings predate service changes, a damper or fan mapping is unverified, a void has not been surveyed after works, or telemetry health is unknown. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - service-connectivity tracing with incident-condition overlay. Decision interval. when spread beyond the apparent room is suspected, before opening or entering a service space and after ventilation or barrier changes. Minimum evidence. a bounded source zone, current service connectivity, identified critical dampers or openings and at least one observation on the suspected pathway. Method. trace documented upstream and downstream connections, show monitored points and gaps, correlate timing without asserting causation, and identify confined-space controls and competent owners. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state hidden-branch risk, airflow assumptions, sensor spacing and any service route derived only from unverified or historical drawings. Safe failure. return SERVICE-SPREAD PATHWAYS INCOMPLETE and require direct inspection and specialist advice.

ANSWER BOUNDARY

Do not predict exact spread, authorise confined-space entry, operate ventilation or dampers, infer absence of contamination from silent sensors or inspect network or research content. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Directs attention to concealed pathways that can move smoke, heat, water or contaminants beyond the room visible to crews.

STAKEHOLDER QUESTION

ER-030

What do verified fixed instruments currently show for heat, smoke, carbon monoxide, specified gases or water conditions, and where are the blind spots?

SENSORS AND TELEMETRY

Proposed evidence. Proposed calibrated temperature, particulate, carbon-monoxide, specified-gas, oxygen, humidity and water-level sensors can provide fixed-point trends; responder instruments remain primary for tactical measurement. Coverage. Named rooms, corridors, stairs, lobbies, plant areas and external points where sensors are commissioned, with location, height, range and blind zones documented. Timing, aggregation and privacy. Raw samples according to sensor response, typically 1 to 30 seconds, with 30-second or 1-minute operational summaries and all original event times retained. Fixed point and zone trend only; no individual exposure, survival or behaviour inference. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Sensors can saturate, drift, be shielded, lose power or communications and cannot represent conditions between points or at crew breathing level. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service instrument users first, then each competent building-system owner and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. direct calibrated responder readings, live commissioned sensor data, calibration and maintenance records, alarm logs, location metadata and incident observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a reading exceeds its sensor-specific operational age, calibration is overdue, health is faulted, or direct responder instruments conflict with building telemetry. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + calculation + estimate - quality-screened multi-sensor time-series comparison. Decision interval. continuously during active monitoring, with reassessment at command-defined intervals and after any abrupt change or sensor loss. Minimum evidence. at least two healthy time points per displayed trend, known sensor location and range, current calibration status and direct responder evidence where a safety decision depends on the reading. Method. align clocks, remove only documented invalid values, display raw and smoothed trends with range and saturation, compare spatially without interpolating across unsupported areas. Decision discipline. Retain inputs, units and formula; label assumptions and range. Uncertainty. show sensor error, age, saturation, missing intervals, spatial separation and disagreement with responder instruments. Safe failure. suppress the derived trend and show CONDITIONS NOT RELIABLY MEASURED when minimum quality is not met. Forecast boundary: do not turn a trend or aligned sequence into a forecast without a verified model, horizon and error bound.

ANSWER BOUNDARY

Do not certify tenability or safety, estimate individual exposure or survival, predict fire growth, issue entry or ventilation commands, or override calibrated responder instruments. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command to distinguish a supported trend from noise, sensor failure or isolated point readings when assessing how the incident is changing.

STAKEHOLDER QUESTION

ER-031

What defensible aggregate occupancy range applied to each affected zone at the incident start time?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous bidirectional doorway or thermal counters may provide zone-level totals where boundaries are closed and validated; only pre-approved irreversible aggregates from other systems may supplement them. Coverage. Public, teaching, office, event and circulation zones with documented counter boundaries; private rooms and uninstrumented routes remain explicit gaps. Timing, aggregation and privacy. Counts every minute with 5-minute summaries; preserve the last healthy reading before alarm time and all reset or outage events. Zone-level ranges with small-count suppression; no names, device identifiers, face images, movement histories or person-level linking. Quality and provenance. Validate each boundary with manual counts, direction/reset checks and an error range; repeat for drift. Align source clocks to the incident timebase; show offset and latency; show uptime, gaps, bypassed routes and resets, and retain only approved aggregates with counter, mapping, software and aggregation provenance. Limits. Tailgating, open doors, counter drift, unmonitored exits and sensor outage can produce cumulative error; occupancy evidence cannot prove who remains. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, direct search teams, Security Control, timetabling and event owners, visitor and contractor record owners, and the building responsible person. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. incident time, current space capacity and use register, timetables and event bookings, authorised aggregate access and visitor counts, commissioned occupancy telemetry and accountable site reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the source does not cover the incident time, a counter has not been reconciled after reset, a timetable or event change is unrecorded, or current room use differs from the controlled register. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + estimate - bounded aggregate occupancy estimation by zone. Decision interval. at initial incident handover, after major evacuation or re-entry changes and when search sectors are revalidated. Minimum evidence. a defined incident time, current zone boundaries, at least two independent aggregate sources or one source with a validated error bound, and clear coverage gaps. Method. reconstruct an occupancy range using last-known counter balance, bookings and accountable reports; suppress small numbers and show lower and upper bounds rather than a precise total. Decision discipline. Retain inputs, units and formula; label assumptions and range. Uncertainty. state counter error, unmonitored doors, data gaps, expected-versus-observed differences and zones with no defensible estimate. Safe failure. return OCCUPANCY ESTIMATE UNAVAILABLE for unsupported zones and rely on search, command and direct accountability processes.

ANSWER BOUNDARY

Do not identify or track anyone, infer attendance from personal devices, declare a zone clear, predict behaviour or survival, or let an estimated count replace search and accountability. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Provides a defensible planning range for search and resource allocation while making clear that aggregate occupancy is not proof of who is present.

STAKEHOLDER QUESTION

ER-032

Which zones are currently declared evacuated, searched, being searched or unconfirmed, and who made each declaration?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous exit counts and zone-level count changes may support the picture; evacuation and search states must come from authorised operational reports. Coverage. Evacuation zones, stair exits, assembly interfaces and search sectors defined for the incident. Timing, aggregation and privacy. Operational status on each authorised report, exit counts every minute and system events at source time; retain superseded statuses in the incident log. Zone and search-sector level only; no responder or occupant identities, routes or individual timestamps. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Exit counts can double-count or miss people, a completed search is time-specific, and no sensor can establish that a zone is clear after conditions or access change. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and authorised search team leaders first, then fire wardens or site evacuation coordinators, Security Control and the building responsible person. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current incident sector board or command log, authorised search reports, evacuation and assembly reports, refuge-call status and aggregate exit evidence. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a status has no observer and time, conditions changed after the report, a zone boundary changed, or a later report conflicts without resolution. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - operational status register with temporal validity and conflict control. Decision interval. on every search or evacuation update and at each command briefing and handover. Minimum evidence. a defined zone or sector, an authorised reporting role, observed time, status vocabulary and any conditions or limitations attached to the report. Method. record evacuated, search in progress, search completed and unconfirmed as distinct states; preserve report provenance and expiry after material change; show sensor evidence only as supporting context. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state whether status is direct, reported or inferred from aggregates, and show when a completed search may no longer represent current occupancy. Safe failure. keep the zone UNCONFIRMED when evidence is missing, stale or conflicting.

ANSWER BOUNDARY

Do not declare a building or zone clear, infer that everyone evacuated, direct search activity, identify occupants or responders, or allow counters to override search-team reports. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives incident command a traceable picture of search and evacuation progress without turning weak occupancy signals into an unsafe clearance claim.

STAKEHOLDER QUESTION

ER-033

Which authorised assistance requests or personal emergency evacuation arrangements are reported by zone, without disclosing identity beyond operational need?

SENSORS AND TELEMETRY

Proposed evidence. Proposed refuge and emergency-assistance call-point activation, connection and fault state can indicate a request location; optional manual status updates must be role-authorised. Coverage. Designated refuges, accessible evacuation points, emergency assistance call points and approved support zones. Timing, aggregation and privacy. Activation and connection events immediately, health heartbeat every minute, and manual updates with observed time and expiry after resolution or relocation. Need category and location only where operationally necessary; restrict names and personal details to emergency-service channels outside the general building service. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. An activation does not describe the person, number, need or current presence; a silent point does not prove no assistance is required, and communications can fail. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service control, then authorised evacuation coordinators, the building responsible person, Security Control and the accessibility or support record owner. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. live assistance-call status, authorised emergency information supplied for the incident, current personal emergency evacuation arrangements where lawfully accessible, and direct reports from the person or assisting staff. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a request has no current contact or status, a person has relocated, a call point is faulted, or a support record is not effective for the present building use or incident. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - assistance-request triage by location, status and communication state. Decision interval. on activation, at each contact or status update and during command reassessments until resolved or handed over. Minimum evidence. an authorised request or direct report, a current location or bounded zone, communication status and an accountable operational owner. Method. display requested assistance, location and last contact without diagnosis; keep active, acknowledged, relocating, resolved and unknown states distinct; apply strict role permissions. Decision discipline. Separate observation from interpretation. Uncertainty. state whether the person remains at the reported point, whether communication is two-way and what details are intentionally unavailable for privacy. Safe failure. show ASSISTANCE REQUEST LOCATION OR STATUS UNCONFIRMED and route the matter through emergency-service command.

ANSWER BOUNDARY

Do not diagnose, predict mobility, survival or behaviour, disclose sensitive personal information beyond operational necessity, identify people in general outputs or issue evacuation instructions. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command prioritise contact and assistance resources while protecting personal information and avoiding assumptions about an individual need.

STAKEHOLDER QUESTION

ER-034

Which refuge or emergency-assistance call points are active, connected, unanswered, faulted or formally closed by command?

SENSORS AND TELEMETRY

Proposed evidence. Proposed call-point controller events can expose activation, call establishment, acknowledgement, line supervision, power and fault state without analysing speech content. Coverage. Every designated refuge, accessible evacuation point and emergency-assistance intercom mapped to the current fire and evacuation strategy. Timing, aggregation and privacy. Events at source time with gateway and retrieval times; health heartbeat every minute; resolution requires an authorised manual status and time. Call-point and zone status only; no audio recording, transcription, speaker identification or content inspection by Talking Abacws. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. A connected line does not prove communication quality or that anyone remains present; a resolved marker can be wrong if not confirmed by the responsible operational role. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and the authorised assistance-call operator, then the fire-safety responsible person, Security Control, system maintainer and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the live assistance-call controller, incident communication log, current point schedule, fault and maintenance records, and direct operational confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a state lacks source time, controller health is unknown, a call has no recent contact update, or the mapped point location has not been checked after alterations. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + diagnosis - call-point state reconciliation and unresolved-call queue. Decision interval. continuously during an active incident and at every command and operator handover. Minimum evidence. current controller health, a uniquely mapped call point, event history and an authorised status owner for each active or unresolved call. Method. order activation, connection, acknowledgement and resolution events; expose faults and communication loss; never infer presence or need from controller state alone. Decision discipline. Root cause needs competent confirmation. Uncertainty. show last confirmed contact, controller and network health, mapping age and whether resolution is technical, reported or directly confirmed. Safe failure. retain the point as UNRESOLVED OR UNAVAILABLE and escalate through incident command.

ANSWER BOUNDARY

Do not monitor, record or interpret speech, identify callers, infer disability or need, mark a request resolved automatically, operate the system or issue instructions through it. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Provides a reliable list of assistance points needing contact or verification instead of treating a technical connection state as a completed rescue.

STAKEHOLDER QUESTION

ER-035

Which current routes, doors, gradients, lifts or platforms could constrain wheelchair, stretcher or assisted movement?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door-width and state records, ramp flood sensors, platform and evacuation-lift status, corridor smoke and water sensing, lighting health and authorised obstruction reports can describe planned accessible routes. Coverage. Accessible external entrances, ramps, wide doors, refuge links, approved lifts and external handover routes. Timing, aggregation and privacy. Asset state on change, environmental evidence every minute and manual route reports with observed time. Route-segment level; no person-level mobility profile, tracking or inferred health information. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Recorded dimensions may not reflect debris or equipment, lift availability can change, and telemetry cannot determine whether a route is clinically or tactically suitable for a particular person. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, ambulance command where relevant, direct crews, Estates access and lift owners, the building responsible person and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current accessible access and evacuation plans, verified dimensions, lift and door status, temporary closures, direct route reports and authorised individual arrangements only when operationally required. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency, treat the answer as stale when a route has not been revalidated after works, a critical asset state is older than 2 minutes, or direct conditions conflict with the plan. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - physical-constraint screening of pre-planned inclusive movement routes. Decision interval. when assistance is requested, before movement and after any route, lift, smoke, water or obstruction change. Minimum evidence. origin and destination zones, current dimensional and asset data for all critical segments, recent condition evidence and command-defined movement requirements. Method. show slope, width, thresholds, doors, lift dependencies and current hazards for each option; do not choose a route or infer individual capability; flag unsupported segments. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state measurement age, debris and smoke gaps, lift confirmation level and where a direct accessibility check is still required. Safe failure. return ACCESSIBLE MOVEMENT ROUTE NOT CONFIRMED and require direct assessment by command and relevant crews.

ANSWER BOUNDARY

Do not infer a person's disability or capability, prescribe movement, certify accessibility or safety, operate lifts or doors, or disclose personal evacuation information beyond need-to-know roles. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps fire and ambulance teams identify physical barriers early and plan the right people and equipment for assisted movement.

STAKEHOLDER QUESTION

ER-036

What aggregate numbers of visitors, contractors, event attendees or other non-routine occupants were expected, and in which zones?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous entrance counters can support aggregate totals; visitor, contractor and event systems may provide permissioned count-only snapshots rather than names. Coverage. Building and event-zone entrances, contractor access points and booked visitor or event areas active at the incident time. Timing, aggregation and privacy. Count snapshots at incident start and every 5 minutes during evacuation; booking and sign-in data effective for the relevant time window. Cohort and zone totals with small-count suppression; no names, contact details, badge identifiers, device identities or movement histories in the general service. Quality and provenance. validate each boundary with manual counts, direction/reset checks and an error range; repeat for drift. Align source clocks to the incident timebase; show offset and latency; show uptime, gaps, bypassed routes and resets, and retain only approved aggregates with counter, mapping, software and aggregation provenance. Limits. People may not sign in or out, events can relocate, counters can miss group movement and booking totals do not prove attendance. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, event and visitor-management owners, contractor control, Security Control and the building responsible person. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current event bookings and attendance reports, contractor and visitor count records, authorised aggregate access data, anonymous counters and accountable organiser reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a list or count does not cover the incident period, event changes are unrecorded, a counter has reset, or reported attendance lacks a responsible owner and observed time. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + estimate - non-routine cohort count reconciliation. Decision interval. at initial accountability assessment, after assembly updates and when the incident affects a booked event or contractor area. Minimum evidence. incident time, current event and contractor schedule, at least one observed or signed-in aggregate and explicit unrecorded-attendance risk. Method. compare expected, signed-in, observed and exited ranges by cohort and zone; suppress identities and show differences without treating them as missing-person counts. Decision discipline. Retain inputs, units and formula; label assumptions and range. Uncertainty. state no-show, walk-in, early departure, shared entrance and counter-error assumptions. Safe failure. show NON-ROUTINE OCCUPANCY UNKNOWN where no defensible aggregate exists.

ANSWER BOUNDARY

Do not disclose names or contact details, identify individuals from badge events, infer who remains, declare anyone missing or safe, or track visitors and contractors. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Alerts command to groups that may not be represented by normal timetable or staff assumptions and may need targeted accountability checks.

STAKEHOLDER QUESTION

ER-037

Which zones must remain 'potentially occupied' because exit, assembly, search or occupancy evidence is incomplete or conflicting?

SENSORS AND TELEMETRY

Proposed evidence. Proposed validated zone and exit counters may provide bounded supporting totals; authorised search, assembly and incident reports remain primary, and environmental presence signals are not used to infer people. Coverage. Incident zones, evacuation routes, exits, assembly interfaces and search sectors with defined evidence boundaries. Timing, aggregation and privacy. Sensor aggregates every minute, reports on each update and a full reconciliation at command-defined reassessment points. Zone-level ranges only with strict small-count suppression and no identity linking across sources. Quality and provenance. Validate each boundary with manual counts, direction/reset checks and an error range; repeat for drift. Align source clocks to the incident timebase; show offset and latency; show uptime, gaps, bypassed routes and resets, and retain only approved aggregates with counter, mapping, software and aggregation provenance. Limits. Different systems count different populations and time windows; residual signals can reflect responders, re-entry or sensor error rather than occupants. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and search-team leaders, then evacuation coordinators, assembly-point leads, Security Control and occupancy-system owners. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current command and search logs, assembly reports, aggregate occupancy and exit data, event or timetable expectations and direct crew observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when sources use different zone boundaries, a search status predates re-entry or change, a counter is unhealthy, or an assembly report lacks time and responsible role. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + estimate - evidence-gap and conflict detection across accountability sources. Decision interval. at each command reassessment, after search completion, after re-entry or relocation and before reducing search attention. Minimum evidence. a common zone map, incident time, latest search state, latest exit or occupancy aggregate and latest assembly report or explicit absence. Method. align sources by zone and time, show supported lower and upper ranges, identify reasons for mismatch and keep conflict visible until an authorised operational report resolves it. Decision discipline. Label assumptions and range; keep conflicts visible. Uncertainty. state coverage and population differences, counter error, possible responder contamination and how long each status remains valid. Safe failure. classify the zone as OCCUPANCY UNCONFIRMED rather than infer presence or absence.

ANSWER BOUNDARY

Do not identify or track people, declare a zone occupied or clear solely from analytics, predict behaviour, assign search tasks or let a model override search-team and command information. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Focuses command attention on accountability gaps that need a report, search or direct check rather than presenting a misleading single occupancy number.

STAKEHOLDER QUESTION

ER-038

Which zones are reported to contain people sheltering in place, and what authorised two-way communication is established with each zone?

SENSORS AND TELEMETRY

Proposed evidence. Proposed help-point and assistance-call state, public-address delivery status and role-approved manual reports can indicate reported shelter locations and communication availability. Coverage. Designated shelter areas, refuges, rooms or zones explicitly reported during the incident; no passive inference from device or network activity. Timing, aggregation and privacy. Reports and communication events at source time, with last successful contact and next reassessment time shown. Zone and communication-status level; personal details remain within emergency-service systems and only minimum necessary information is exposed. Quality and provenance. verify asset-location mapping and end-to-end states; binary contacts need functional tests, not calibration. Record clock offset/latency, heartbeat, uptime, gaps, stuck or chattering states and mapping drift; retain raw-event, firmware and transformation provenance. Limits. Message delivery does not prove receipt or understanding, a call can drop, people can relocate and no system can infer sheltering from silence or occupancy telemetry. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, police or ambulance command where relevant, authorised communication operators, the building responsible person and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. direct emergency-service reports, authorised assistance-call and incident communication logs, current shelter or evacuation strategy and accountable site reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when there has been no contact within the command-defined reassessment interval, location is not reconfirmed after relocation, or communication-system health is unknown. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - reported shelter-location and communication-status register. Decision interval. on first report, every contact attempt or status change and at each command briefing until resolved. Minimum evidence. an authorised report, a bounded location, last contact time, communication channel and accountable command owner. Method. record reported location, approximate aggregate where authorised, communication status and stated limitations; do not monitor content or infer intent; retain superseded locations with time. Decision discipline. Separate observation from interpretation. Uncertainty. state whether location and numbers are direct, reported or unknown, and whether communication was sent, connected or acknowledged. Safe failure. show SHELTER LOCATION OR CONTACT UNCONFIRMED and escalate through incident command.

ANSWER BOUNDARY

Do not infer who is sheltering, monitor conversations, identify people, predict behaviour or survival, issue public instructions, or treat message delivery as confirmation of safety. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command maintain contact responsibility and location awareness for reported shelter groups without creating covert surveillance or false reassurance.

STAKEHOLDER QUESTION

ER-039

After relocation, partial reopening or controlled re-entry, which zones have a new occupancy state and which earlier status is no longer valid?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous entrance and exit counts, zone occupancy counters and access-event aggregates can support change detection when re-entry points are defined. Coverage. Only zones and controlled access points included in the incident relocation or re-entry plan. Timing, aggregation and privacy. Counts every minute, authorised movement phases by effective time and status updates on every command decision. Zone-level net change and ranges; no individual identity, badge trail or movement history. Quality and provenance. validate each boundary with manual counts, direction/reset checks and an error range; repeat for drift. Align source clocks to the incident timebase; show offset and latency; show uptime, gaps, bypassed routes and resets, and retain only approved aggregates with counter, mapping, software and aggregation provenance. Limits. Uncontrolled doors, counter drift, responder movements and mixed flows can make net counts unreliable; analytics cannot distinguish authorised from unauthorised movement without direct operational records. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, police or security control for access boundaries, evacuation coordinators, search-team leaders and the building responsible person. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current command decisions, relocation and re-entry logs, zone search status, authorised aggregate access counts, anonymous counters and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when zone boundaries or movement phases changed, counters were not reset or reconciled, or a status predates the latest re-entry or search event. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + estimate - phase-aware occupancy-status reconciliation. Decision interval. at each relocation or re-entry phase, after a search update and before any handover or relaxation of controls. Minimum evidence. a command-defined movement phase, controlled zone boundaries, latest search status and at least one current aggregate or direct report. Method. compare pre-phase and post-phase ranges, separate responder and public movement where operational reports allow, and keep zones unconfirmed when flow cannot be attributed. Decision discipline. Label assumptions and range; keep conflicts visible. Uncertainty. state counter reset quality, uncontrolled access risk, responder contamination and the time window represented. Safe failure. retain CHANGED STATUS UNCONFIRMED and require direct accountability and search information.

ANSWER BOUNDARY

Do not authorise re-entry, identify individuals, infer compliance, declare zones clear, predict movement or let access analytics override incident-command restrictions. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents earlier evacuation or search status being reused after people have moved and conditions have changed.

STAKEHOLDER QUESTION

ER-040

Where are the largest unresolved differences between expected occupancy, exit counts, assembly reports and completed search records?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous occupancy and exit counts can contribute bounded totals; the analysis also uses authorised aggregate booking, visitor, assembly and search-status records. Coverage. All command-defined zones, exits, assembly areas and search sectors for which at least one evidence source exists. Timing, aggregation and privacy. Recalculate at each command reassessment and on any material source update; preserve the incident-time baseline and all subsequent phases. Zone and cohort ranges with small-count suppression; no identity resolution or cross-source person matching. Quality and provenance. validate each boundary with manual counts, direction/reset checks and an error range; repeat for drift. Align source clocks to the incident timebase; show offset and latency; show uptime, gaps, bypassed routes and resets, and retain only approved aggregates with counter, mapping, software and aggregation provenance. Limits. Source populations, boundaries and clocks differ, duplicates are possible, and a numerical gap cannot be interpreted as a number of people missing or remaining. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and accountability leads, then search-team leaders, assembly-point leads, Security Control, event and occupancy data owners. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current command log, expected occupancy ranges, exit counts, assembly reports, search status, authorised visitor and event aggregates and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when sources use unreconciled time windows or zones, a search report predates re-entry, a count is unhealthy, or an assembly report has not been refreshed within the command reassessment interval. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. calculation + comparison + estimate - multi-source accountability gap analysis with bounded differences. Decision interval. at command-defined accountability reassessments, after major search completion and after relocation or re-entry. Minimum evidence. common zone and time definitions, at least three relevant source classes or an explicit missing-source statement, and current search status. Method. normalise source scope, calculate only defensible lower and upper differences, rank gaps by evidence weakness and operational significance, and show the cause of each mismatch. Decision discipline. Retain inputs, units and formula; label assumptions and range; keep conflicts visible. Uncertainty. state duplicate and omission risk, source coverage, counter error, boundary mismatch and why the gap is not a person count. Safe failure. show ACCOUNTABILITY GAP NOT QUANTIFIABLE where assumptions cannot be bounded.

ANSWER BOUNDARY

Do not identify or track individuals, declare anyone missing, trapped, safe or deceased, predict survival or behaviour, or use analytics to replace search and formal accountability. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command decide which zones or source owners need the next verification effort without presenting a false exact missing-person figure.

STAKEHOLDER QUESTION

ER-041

Which alarm device or zone has the earliest defensible state change, and what clock or mapping uncertainty affects that claim?

SENSORS AND TELEMETRY

Proposed evidence. Proposed read-only fire-alarm panel and gateway interfaces can expose detector, manual-call-point, interface and zone events together with device status, clock health and gateway latency. Coverage. Every commissioned fire-alarm loop, zone, device and interface mapped to the current panel configuration. Timing, aggregation and privacy. Preserve original panel event time, panel receipt sequence, gateway receipt and responder-service retrieval time; do not replace source ordering with display time. Device and zone event level; no association with individual occupants or callers. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Panel clocks can drift, buffered gateways can reorder delivery, devices can be mislabelled and the first recorded activation may not be the first physical event or origin. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct emergency-service interpretation, then the fire-alarm responsible person, panel maintainer, Estates fire systems owner and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the authoritative panel event log, controller configuration and zone schedule, clock and maintenance records, incident reports and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the panel clock offset is unknown, logs are incomplete, device mapping predates reprogramming or alteration, or an event was retrieved from a cached rather than current source without disclosure. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + diagnosis - event-order reconstruction with clock-quality assessment. Decision interval. at initial alarm handover, when origin or spread chronology is revalidated and during post-incident evidence preservation. Minimum evidence. the unaltered event sequence, panel and gateway clock status, current device mapping and at least one independent incident observation or system event. Method. retain source order, quantify known clock offset and delivery delay, identify the earliest reliably ordered event and list events whose ordering overlaps within uncertainty. Decision discipline. Retain inputs, units and formula; root cause needs competent confirmation. Uncertainty. state whether first means first recorded, first received or earliest within a time range; show missing logs, clock drift and mapping confidence. Safe failure. return FIRST ALARM EVENT NOT DETERMINABLE rather than select an event from incomplete or unsynchronised records.

ANSWER BOUNDARY

Do not determine fire origin or cause, identify who operated a call point, alter or acknowledge alarms, certify panel accuracy or let chronology override incident-command interpretation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives command and later investigators an honest earliest-event window without presenting a panel timestamp as proof of physical origin.

STAKEHOLDER QUESTION

ER-042

What time-aligned sequence is visible across alarm devices, zones and interfaces, including repeats, resets, faults and communication delays?

SENSORS AND TELEMETRY

Proposed evidence. Proposed read-only interfaces can collect fire-alarm, sprinkler, smoke-control, lift recall, door-release, emergency-lighting and power events while preserving each source identifier and clock. Coverage. Only commissioned life-safety systems and interfaces whose asset-to-zone mapping and permission have been verified. Timing, aggregation and privacy. Event-level source times, receipt times and retrieval times retained; clock offsets and periods of loss recorded explicitly. Asset and zone sequence; no person-level inference or linking to access events. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Different controllers have different clocks and event semantics, network buffering can delay records and missing interface events can create an apparent sequence that never occurred. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, each competent life-safety system owner, the fire-safety responsible person, Estates Control and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. authoritative controller logs, cause-and-effect matrix, interface configuration, clock-synchronisation records, incident reports and direct crew observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an interface has unknown health, controller time is not reconciled, configuration predates change, or event extraction is partial or transformed without provenance. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + comparison - multi-system temporal alignment and sequence visualisation. Decision interval. during command reassessment when system progression matters and during evidence preservation and handover. Minimum evidence. unaltered logs from each included source, known or bounded clock offsets, current interface mapping and explicit indication of missing systems. Method. normalise event semantics without replacing source text, align by bounded clock ranges, group near-simultaneous events and show causal design links separately from observed order. Decision discipline. Retain inputs, units and formula; keep conflicts visible. Uncertainty. display time windows, missing intervals, source health and whether a relationship is designed, merely sequential or directly observed. Safe failure. show SEQUENCE PARTIAL OR UNRESOLVED and omit derived ordering where clock ranges overlap materially.

ANSWER BOUNDARY

Do not infer cause, operate or reset systems, identify individuals, certify that designed cause-and-effect occurred, or present temporal correlation as proof of causation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports a shared operational timeline and reveals whether apparently inconsistent system behaviour may instead reflect clock, mapping or interface problems.

STAKEHOLDER QUESTION

ER-043

What does the sprinkler or water-mist system indicate about activation, valves, pumps, tanks, flow and pressure, and which states still need physical confirmation?

SENSORS AND TELEMETRY

Proposed evidence. Proposed flow-switch, pressure, tank-level, pump run or fault, valve tamper and local controller signals can describe the suppression water system at defined points. Coverage. Each sprinkler or water-mist zone, control valve, pump set, tank, monitored flow point and interface serving Abacws. Timing, aggregation and privacy. Flow, pump and valve changes as events; pressure and tank level every 10 to 30 seconds; show health and last inspection status. System, zone and asset level; no inference about occupants or fire conditions beyond the indicated activation. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. A flow switch can be triggered by testing or damage, pressure is local and transient, remote valve indication may be wrong, and telemetry cannot prove discharge effectiveness or water availability at every head. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and fire-service water or suppression specialists, then the sprinkler responsible person, Estates water and fire-system owners and maintenance contractor. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. authoritative suppression controller and pump logs, valve and zone schedule, recent inspection and test records, impairment register, water-supply information and direct gauge or valve observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a critical signal is older than 1 minute, maintenance or testing status is unresolved, a valve mapping predates change, or system health and direct observation conflict. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + diagnosis - suppression-system state reconciliation by zone and dependency. Decision interval. at initial system handover, after any flow, pressure or pump event and throughout water-supply and recovery decisions. Minimum evidence. identified system and zone, current controller health, valve and pump state, at least one pressure or flow observation and an accountable competent owner. Method. separate alarm, flow, pump, valve and supply evidence; display local readings and test states; compare against design intent without estimating extinguishing performance. Decision discipline. Root cause needs competent confirmation. Uncertainty. show sensor location, range, age, test-mode risk, unmonitored valves and whether operation is confirmed by direct observation. Safe failure. state SUPPRESSION OPERATION OR WATER STATE NOT VERIFIED and rely on direct fire-service checks.

ANSWER BOUNDARY

Do not operate pumps or valves, certify water supply or suppression effectiveness, infer fire control, recommend tactics or let telemetry replace fire-service inspection. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives crews a rapid, source-linked picture of whether water systems appear to have operated and which supply or valve states need physical confirmation.

STAKEHOLDER QUESTION

ER-044

What is the intended incident function and current indicated state of smoke-control fans, dampers, vents and pressurisation systems?

SENSORS AND TELEMETRY

Proposed evidence. Proposed fan run and trip feedback, damper and vent position, motor current, pressure differential and controller mode can report selected smoke-control states. Coverage. Each smoke-control zone, fan, damper, vent, pressurised stair or lobby and control interface documented for Abacws. Timing, aggregation and privacy. Commands and state changes as events; pressure every 5 to 30 seconds; health and manual-mode status with every retrieval. Asset and smoke-control zone level; no person or movement inference. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Command and feedback can disagree, a damper position does not prove airflow, pressure sensors are local and door opening changes performance, while fire damage can invalidate design assumptions. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then the smoke-control responsible person, fire-safety management, Estates mechanical engineering and the system maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire strategy and smoke-control cause-and-effect, controller logs, fan and damper schedules, commissioning and maintenance records, impairment log and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when design records predate alteration, a controller is in manual, test or fault without explanation, pressure is older than 30 seconds, or feedback lacks current health. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - design-intent-to-current-state reconciliation. Decision interval. at initial life-safety handover, after any mode, fan, damper, vent or door change and before relying on a protected route. Minimum evidence. current design intent, uniquely identified assets, command and feedback state, pressure or airflow evidence where relevant and a competent system owner. Method. display designed action, current command, physical feedback and measured pressure separately; expose mismatches, manual interventions and unmonitored components. Decision discipline. Root cause needs competent confirmation. Uncertainty. state local sensor limitations, door-state dependencies, unverified airflow and whether fire conditions may have exceeded design basis. Safe failure. return SMOKE-CONTROL PERFORMANCE NOT ESTABLISHED and require direct incident and competent-system assessment.

ANSWER BOUNDARY

Do not operate smoke control, certify pressurisation or tenability, predict smoke movement, recommend ventilation tactics or treat a run signal as proof of effective airflow. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps commanders distinguish what the system was designed to do from what controllers and field conditions currently indicate.

STAKEHOLDER QUESTION

ER-045

Which fire-alarm zones, loops, devices or interfaces are disabled, isolated, in test or faulted, who authorised the condition, and since when?

SENSORS AND TELEMETRY

Proposed evidence. Proposed read-only panel interfaces can expose fault, disablement, isolation, test, contamination and communication-loss states with original device and operator-role records where available. Coverage. All panel loops, zones, devices and life-safety interfaces included in the current fire-alarm configuration. Timing, aggregation and privacy. State changes as events with effective, source and retrieval times; panel health heartbeat every 30 seconds. Asset and zone level; do not reveal individual operator identities except through authorised formal records outside the general display. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Panel state may not reflect a physically removed detector, undocumented local bypass or field damage, and missing events can obscure when an impairment began. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, the fire-alarm responsible person, fire-safety management, Estates fire systems owner and authorised maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. authoritative panel status and logs, current impairment and permit register, maintenance records, cause-and-effect configuration, direct inspection and accountable site handover. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an impairment has no effective and reassessment time, panel health is unknown, mapping predates change, or a manual report conflicts with panel state. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + diagnosis - impairment register reconciliation with live panel status. Decision interval. at initial handover, on every impairment state change and at each command and recovery reassessment. Minimum evidence. current panel health, complete active state list, asset-to-zone mapping, impairment owner and start or last-confirmed time. Method. group by fault, disabled, isolated and test states; show operational coverage impact and unresolved discrepancies; preserve source wording and history. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unknown start times, unmonitored field conditions, partial extraction and whether the impairment affects detection, warning or an interface. Safe failure. treat the affected coverage as UNAVAILABLE OR IMPAIRED until direct confirmation.

ANSWER BOUNDARY

Do not enable, disable, reset or acknowledge devices, validate the authority of a past action, identify individual operators or certify remaining alarm coverage. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command to recognise where expected detection or interfaces cannot be relied on and to request the correct competent confirmation.

STAKEHOLDER QUESTION

ER-046

What is known about emergency-lighting and illuminated exit-sign coverage on the routes under consideration, including unmonitored sections?

SENSORS AND TELEMETRY

Proposed evidence. Proposed central-battery controller status, monitored luminaire and sign faults, circuit power, battery state and selected fixed illuminance sensors can indicate parts of the emergency-lighting system. Coverage. Pre-defined emergency routes, stairs, lobbies, final exits, refuge links and external transitions with mapped monitored assets. Timing, aggregation and privacy. Power and fault changes as events, battery status every minute and illuminance every 30 seconds where installed. Route segment and asset level; no camera-based observation of people or movement. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Most luminaires may not be individually monitored, local smoke or damage can obscure light, illuminance sensors represent one point and controller health cannot prove visibility along the route. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then the emergency-lighting responsible person, Estates electrical owner, fire-safety management and maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current emergency-lighting and sign schedule, central controller status, inspection and duration-test records, circuit information, impairments and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a test or inspection is overdue, asset mapping predates alteration, battery data is older than 5 minutes during outage, or direct observations conflict with controller state. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - route-level emergency-lighting evidence summary. Decision interval. after normal power loss, before relying on a route and after any lighting, smoke or structural change. Minimum evidence. identified route, current power and controller health, latest inspection or test status and direct confirmation for critical segments not individually monitored. Method. map known healthy, faulted and unmonitored assets to route segments, compare fixed illuminance only at its measured location and show coverage gaps. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unmonitored luminaires, battery-duration assumptions, smoke obscuration and sensor-location limits. Safe failure. show EMERGENCY LIGHTING NOT VERIFIED FOR ROUTE and require direct lighting and visibility assessment.

ANSWER BOUNDARY

Do not certify route visibility or duration, operate lighting circuits, direct route use, infer safe egress from controller status or replace direct crew observation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps crews distinguish a monitored system fault from unverified local darkness before committing to a route during power or smoke conditions.

STAKEHOLDER QUESTION

ER-047

Which fire doors, shutters or hold-open devices have released, remain open, are faulted or are reported damaged at critical compartment lines?

SENSORS AND TELEMETRY

Proposed evidence. Proposed door and shutter position contacts, hold-open release and power state, closer-cycle monitoring and obstruction alarms can indicate selected asset states. Coverage. Critical doors and shutters on compartment boundaries, protected stairs, lobbies, corridors and high-risk rooms included in the fire strategy. Timing, aggregation and privacy. Position and release changes as events, health heartbeat every minute and manual damage reports with observed time. Door or shutter asset level; no identity or movement history from openings. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. A closed signal does not prove latching, seal or integrity, repeated openings may be responders, and an obstruction or physical damage can be invisible to a simple contact. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then fire-safety management, Estates building owner, access-control owner and maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire-door and shutter schedule, live position and release state, inspection and maintenance records, impairment log and direct reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when inspection is overdue, asset mapping predates alteration, a state is older than 2 minutes during active change, or direct observation conflicts with telemetry. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + comparison + diagnosis - critical barrier-state reconciliation. Decision interval. at initial compartment assessment, after alarm release, on any door or shutter change and during recovery inspection. Minimum evidence. verified asset identity and boundary role, current indication, latest inspection or damage report and direct confirmation before tactical reliance. Method. show intended function, current indicated state, reported damage and inspection condition separately; flag unmonitored or disputed openings. Decision discipline. Root cause needs competent confirmation; keep conflicts visible. Uncertainty. state contact type, latching and integrity blind spots, observation age and whether an opening may be operationally held. Safe failure. treat the opening as BARRIER STATUS UNKNOWN and require direct assessment.

ANSWER BOUNDARY

Do not close, release or control doors or shutters, track users, certify fire resistance or compartmentation, or promise that an indicated closed barrier will contain spread. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Identifies openings that may compromise intended protection and need rapid visual confirmation without overstating what door telemetry proves.

STAKEHOLDER QUESTION

ER-048

Where are the relevant dry or wet riser inlets and landing valves, and what test, valve, pump or pressure evidence is current?

SENSORS AND TELEMETRY

Proposed evidence. Proposed riser pressure, pump state, tank level, valve position and cabinet access sensors can provide selected evidence on a wet or monitored dry riser. Coverage. External inlets, pumps and tanks, each landing valve, relevant isolation valves and associated fire-fighting shafts. Timing, aggregation and privacy. Pressure every 10 to 30 seconds, valve and pump changes as events, and inspection or flow-test records by effective and test time. Riser, inlet and landing-valve asset level; no responder tracking. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Pressure is location-specific, dry risers may have no live water indication, cabinet state does not prove valve condition and telemetry cannot establish flow performance at the branch in use. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and fire-service water officers, then the riser responsible person, Estates water and fire-system owners and maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current riser and landing-valve schedule, verified drawings, inspection and pressure or flow-test results, impairment log, live pump or pressure data and direct gauge observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the last test is overdue, mapping predates alteration, a pressure value is older than 1 minute, or an inlet or valve condition is not directly confirmed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - riser location and readiness evidence retrieval. Decision interval. during water-supply planning, before reliance on a riser and after any valve, pump or pressure change. Minimum evidence. verified inlet and landing-valve locations, current system type, latest test status, impairment state and direct local confirmation for the point to be used. Method. map the service path, present design and test data separately from current indication, and show unknown valves, unavailable pressure and access constraints. Decision discipline. Root cause needs competent confirmation. Uncertainty. state test age, sensor location, dry-riser limitations, gauge accuracy and any unverified branch or isolation. Safe failure. return RISER OR LANDING-VALVE READINESS NOT VERIFIED and rely on fire-service checks.

ANSWER BOUNDARY

Do not operate valves or pumps, certify pressure or water delivery, direct hose deployment or let a historical test substitute for current fire-service verification. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Reduces time spent searching for inlets and valves while keeping current water readiness subject to direct fire-service assessment.

STAKEHOLDER QUESTION

ER-049

Has any gaseous, foam, kitchen or other specialist suppression system entered alarm, pre-discharge, discharge, abort, fault or isolation state?

SENSORS AND TELEMETRY

Proposed evidence. Proposed specialist suppression controller interfaces can expose alarm, confirmed detection, pre-discharge timer, discharge, abort, agent-pressure, isolation and fault states. Coverage. Every registered gaseous, foam, wet-chemical, kitchen or process suppression zone and its warning, interlock and ventilation interfaces. Timing, aggregation and privacy. Controller events at source time, gateway receipt and retrieval time; pressure or agent condition at system-specific intervals. System and protected-zone level; no person identity or inference about presence inside the zone. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Controller indication may not prove agent release or concentration, local abort or manual actions may be missing, and damaged sensors or piping can invalidate expected performance. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and specialist responders, then the suppression-system responsible person, fire-safety management, process owner, Estates engineering and maintainer. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. authoritative controller log, current protected-zone and cause-and-effect information, agent and hazard data, maintenance and impairment records, permit status and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when controller health is unknown, mapping predates change, pressure or agent data is overdue, or manual intervention lacks time and accountable owner. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + diagnosis - specialist suppression state and protected-zone reconciliation. Decision interval. immediately on any pre-discharge or discharge event, at hazard handover and throughout access, ventilation and recovery reassessment. Minimum evidence. identified system and zone, current controller state and health, agent or hazard information, relevant interlock state and direct confirmation where exposure or entry decisions depend on it. Method. display every state transition and timer with source time, distinguish commanded from confirmed discharge and show unresolved interlock, isolation and fault conditions. Decision discipline. Root cause needs competent confirmation. Uncertainty. state agent-release confirmation limits, concentration unknowns, unmonitored spaces, clock quality and possible manual actions. Safe failure. treat the protected area as SPECIALIST SUPPRESSION STATE UNKNOWN and escalate through incident command.

ANSWER BOUNDARY

Do not start, abort, reset or isolate suppression, certify agent concentration or area safety, predict occupant outcomes or issue entry and ventilation instructions. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Warns responders about pre-discharge or discharged systems and the specific competent information needed before the protected area is approached.

STAKEHOLDER QUESTION

ER-050

Which life-safety telemetry and interfaces are currently healthy enough to display, and where must the service show unavailable or conflicting evidence?

SENSORS AND TELEMETRY

Proposed evidence. Read-only health interfaces may expose heartbeat, power, communications, device disablement, mapping version, calibration or test status, latency, clock offset and data loss for selected life-safety panels and gateways. They must not acknowledge, reset, silence, isolate or control any system. Coverage. Only commissioned interfaces and mapped devices; local-only indicators, damaged field wiring and unintegrated systems remain explicit gaps. Timing, aggregation and privacy. Event-driven faults and state changes, heartbeat at the approved interval, and short trend summaries for latency or packet loss; asset-level data only. Quality and provenance. Validate interface mapping end-to-end, compare against the authoritative panel, retain original event IDs, source and retrieval times, clock offset, gateway version, uptime, gaps, resets and transformations. Limits. A healthy interface does not prove detector coverage, fire-system performance or field conditions; an unhealthy feed may leave the underlying panel operational but unavailable to the service. Deployment status. Proposed; hide or mark each feed UNAVAILABLE unless commissioning, permission and current health are verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational reliance, then each competent system owner, Estates digital and network operations, fire-safety management and information governance. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. commissioning and acceptance records, live health and clock status, calibration and maintenance history, asset-to-zone mapping, permissions and direct system-panel checks. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when health is beyond the system-specific heartbeat, calibration or commissioning has expired, permissions are absent, or a source and direct panel state conflict. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Diagnosis + comparison - responder-facing evidence-health assessment, not life-safety system diagnosis. Decision interval. On initial connection, after any fault or restart, at each command handover and before displaying a system state as current. Minimum evidence. A mapped source, successful end-to-end test, current heartbeat, known clock offset, acceptable latency or gap status and permission for the receiving role. Method. Evaluate each feed independently, compare with the authoritative panel where available, and label data confirmed, delayed, conflicting or unavailable. Decision discipline. Root cause and system fitness require the competent system owner. Uncertainty. Show unmonitored assets, interface-only coverage, mapping age, clock error and data loss. Safe failure. Suppress the affected state and display TELEMETRY NOT RELIABLE rather than presenting stale or partial data.

ANSWER BOUNDARY

Do not diagnose the life-safety system, suppress or acknowledge panel events, certify system performance or display stale interface data as current. Experimental telemetry never overrides the authoritative panel, direct observation or incident command.

WHY THIS IS USEFUL

Prevents command from seeing a polished but misleading dashboard when mappings, clocks, interfaces or permissions are incomplete.

STAKEHOLDER QUESTION

ER-051

Where is the current command rendezvous or command-post location, when was it confirmed, and which role-approved channel reaches the site liaison?

SENSORS AND TELEMETRY

Proposed evidence. Proposed external smoke, wind, noise, lighting, anonymous area-occupancy and local radio-infrastructure health can describe the nominated area; the command location itself comes from an authorised operational report. Coverage. Pre-defined rendezvous and command-post candidates, adjacent access routes and fixed communications infrastructure. Timing, aggregation and privacy. Command-location updates as events, environmental summaries every minute and communications-health heartbeat every 30 seconds. Zone and channel-status level; no tracking of command staff, devices or radio users. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timeline and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Infrastructure health does not prove usable communications at every position, conditions can change quickly and a nominated point can be moved without the site system receiving an update. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service control first, then the authorised site liaison, Security Control, Estates communications owner and relevant agency liaison officers. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current command log, authorised rendezvous decision, role and contact directory, radio or telephony infrastructure status, access constraints and direct on-scene confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the location or channel has not been reconfirmed after command movement, a contact role has changed shift, or infrastructure state is older than 2 minutes. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - authorised command-location and contact-channel publication. Decision interval. at first command establishment, on every move or channel change and at each multi-agency handover. Minimum evidence. an explicit command-authorised location, effective time, responsible liaison role and a tested or directly confirmed contact method. Method. publish one current operational location and channel with superseded values retained in the log; show environmental and infrastructure evidence separately from the command decision. Decision discipline. Separate observation from interpretation. Uncertainty. state last confirmation time, channel test status, sensor distance and any agency-specific access restriction. Safe failure. show COMMAND RENDEZVOUS OR CHANNEL NOT CONFIRMED and refer users to emergency-service control.

ANSWER BOUNDARY

Do not choose or move the command post, create operational channels, disclose restricted frequencies or credentials, track users or represent infrastructure health as proof of communication. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives arriving agencies and site representatives one current command location and liaison channel, reducing missed handovers and parallel contact routes.

STAKEHOLDER QUESTION

ER-052

Which authorised site representatives and competent system owners are available now, and what can each person legitimately confirm?

SENSORS AND TELEMETRY

Proposed evidence. Proposed duty-system acknowledgements and role availability status can indicate whether a responsible function has accepted contact; no location tracking is required. Coverage. Fire safety, Security Control, Estates duty management, electrical, mechanical, lift, laboratory, hazardous-material, accessibility and information-system roles relevant to Abacws. Timing, aggregation and privacy. Duty rosters by effective shift; acknowledgement and call status as events; last confirmation displayed. Role and contact-channel level; names and personal contact details shown only to authorised emergency-service users when necessary. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. A roster can be wrong, a phone can be unanswered and an acknowledgement does not prove competence for the exact asset or authority to act. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, emergency-service liaison, the university duty manager, Security Control, Estates operations and each professional duty-roster owner. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current role and competence register, duty and escalation rosters, authorised contact directory, acknowledgement log and direct confirmation from the responsible function. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the shift has changed, competence or delegation has expired, the person has not acknowledged, or contact details conflict across current records. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation - role-to-incident need matching and availability confirmation. Decision interval. at mobilisation, initial handover, specialist escalation and every shift or command handover. Minimum evidence. a defined role requirement, current roster, authorised contact route, competence or delegation scope and a current acknowledgement or explicit no-response state. Method. match incident information needs to accountable roles, display primary and escalation functions, preserve privacy and distinguish listed, contacted, acknowledged and on-scene states. Decision discipline. Separate observation from interpretation. Uncertainty. state whether authority and competence are documented, whether availability is confirmed and whether the role covers the exact system or area. Safe failure. show COMPETENT OWNER NOT CONFIRMED and escalate through the university duty manager and incident command.

ANSWER BOUNDARY

Do not track people, infer availability from device location, disclose personal data beyond operational need, validate professional competence beyond current records or authorise anyone to act. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Speeds access to the person who can explain or verify a system without presenting a generic contact as a competent decision-maker.

STAKEHOLDER QUESTION

ER-053

What concise, time-stamped site handover is available for incident location, hazards, occupancy, systems, access and actions already taken?

SENSORS AND TELEMETRY

Proposed evidence. Read-only extracts from authorised incident, alarm, access, hazard, occupancy and life-safety sources may populate a source-linked handover template; authorised site observations remain clearly attributed manual entries. No free-text fact may be generated without a cited source. Coverage. The agreed handover headings only: location, access, hazards, occupancy and search status, system states, actions, owners, conflicts and missing evidence. Timing, aggregation and privacy. Refresh on material source change and freeze a time-stamped snapshot for each handover. Show role and contact channel rather than unnecessary personal data; suppress restricted detail outside approved roles. Quality and provenance. Preserve source wording, event IDs, effective, observed, source-event and retrieved times, clock offsets, template version, permissions and all later corrections. Limits. Automated extraction can omit context, duplicate events or carry stale mappings; the summary cannot know unreported actions and is never a complete operational briefing. Deployment status. Proposed; return INCOMPLETE HANDOVER when any critical heading lacks current, permissioned and traceable evidence.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command determines operational use; the authorised site liaison owns the site handover, supported by fire safety, Security Control, Estates and relevant specialist owners. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current incident log, direct site reports, alarm and life-safety panel status, hazard registers, occupancy ranges, active isolations or impairments and authorised action records. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when any statement lacks an owner and time, source health is unknown, a later report conflicts, or the handover has not been refreshed after a material change. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + comparison - source-linked handover compilation. Decision interval. At first site liaison, each command or shift handover, and after a material change to location, hazards, occupancy, access or system state. Minimum evidence. For every critical heading, one current source, owner, time basis, limitation and any conflict; manual actions require an authorised reporter and observed time. Method. Populate the fixed template from cited source records, preserve source wording, separate facts from estimates, and place unresolved items in a visible gap list. Decision discipline. The generated pack supports, but never replaces, verbal handover and command confirmation. Uncertainty. Show omitted headings, stale mappings, clock differences and unreported actions. Safe failure. Issue INCOMPLETE HANDOVER with open items instead of filling gaps.

ANSWER BOUNDARY

Do not issue operational commands, omit known conflicts, invent missing facts, expose restricted information to the wrong role or present the compiled pack as a complete briefing without human handover.

WHY THIS IS USEFUL

Provides a disciplined briefing in which every critical fact has a source, time, owner and visible limitation during a high-pressure handover.

STAKEHOLDER QUESTION

ER-054

Where are known radio or mobile communication weak areas, and what is the present health of fixed repeaters, distributed antenna or leaky-feeder systems?

SENSORS AND TELEMETRY

Proposed evidence. Proposed infrastructure heartbeat, power, fault and signal-level probes can report repeater and distributed-antenna health; periodic radio surveys can map baseline coverage without recording communications. Coverage. Basements, stairs, fire-fighting shafts, plant rooms, deep internal zones, roof areas and external command candidates. Timing, aggregation and privacy. Infrastructure health every 30 seconds; survey maps by test date and configuration; direct crew reports at observed time. Area and infrastructure level; no interception, recording, transcription, caller identity or individual radio-location data. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timescale and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Baseline surveys do not represent fire damage, crowding, doors, handset types or all agency frequencies; a healthy repeater may still provide poor operational coverage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service communications specialists, then Estates communications owner, Security Control and authorised network maintainers. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current radio survey, repeater and power status, maintenance and fault records, approved frequency and system scope, incident reports and direct radio checks. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a survey predates building or system change, infrastructure status is older than 1 minute, or crews report failure inconsistent with the baseline. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - communications coverage and infrastructure-health overlay. Decision interval. during command-post selection, before deployment to known difficult areas and whenever crews report communication failure. Minimum evidence. current infrastructure health, survey scope and date, direct operational test or report for the relevant area and clear indication of agency and frequency applicability. Method. show baseline weak zones, current infrastructure faults and live reports separately; never infer individual radio positions or monitor content. Decision discipline. Root cause needs competent confirmation. Uncertainty. state survey conditions, frequency and handset scope, untested areas, current building changes and conflict with field reports. Safe failure. show COMMUNICATION COVERAGE UNCONFIRMED and rely on emergency-service communications procedures.

ANSWER BOUNDARY

Do not monitor or record communications, disclose restricted frequencies, track handsets, promise coverage or direct deployment based solely on a survey map. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command anticipate where communication checks or alternative procedures may be needed before teams enter known difficult areas.

STAKEHOLDER QUESTION

ER-055

Which public-address, voice-alarm or mass-notification zones received the latest authorised message, and which delivery paths reported faults?

SENSORS AND TELEMETRY

Proposed evidence. Proposed controller logs can expose selected zone activation, message identifier, amplifier and circuit health, end-of-line supervision and fault state without capturing audience responses. Coverage. Every mapped public-address, voice-alarm and approved mass-notification zone and its critical distribution interfaces. Timing, aggregation and privacy. Authorised message issue and controller events at source time, delivery status on completion and system health every minute. Message and zone status; no recipient identities, device tracking, microphone monitoring or speech analysis. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. A sent or electrically delivered message does not prove audibility, receipt, understanding or compliance, especially in noise, smoke or hearing-accessibility contexts. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and the authorised warning issuer, then the voice-alarm or notification system owner, fire-safety management, Security Control and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the authorised message and issue log, controller zone and fault logs, current zone mapping, system test and impairment records and direct reports of audibility or failure. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the message has been superseded, zone mapping predates change, controller health is unknown or a fault affects distribution without acknowledged operational handling. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + diagnosis - authorised warning-distribution status and fault reconciliation. Decision interval. after each authorised message, on any distribution fault and during command reassessment of public information. Minimum evidence. message identifier and issuer, effective time, targeted zones, controller outcome, current mapping and explicit limitations on receipt and understanding. Method. show targeted, initiated, electrically completed, faulted and unconfirmed zones separately; link direct audibility reports without inferring audience response. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unmonitored loudspeakers, ambient noise, language and accessibility limitations, device delivery semantics and any network delay. Safe failure. show MESSAGE DELIVERY NOT CONFIRMED for affected zones and refer further warning decisions to incident command.

ANSWER BOUNDARY

Do not issue or edit messages, infer that people heard, understood or complied, identify recipients, monitor speech or allow the system to replace authorised warning procedures. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps command see where an authorised warning may not have been distributed as intended and where another verified communication route may be needed.

STAKEHOLDER QUESTION

ER-056

Are incident command, site teams and building systems using the same names for entrances, floors, zones, sectors, assembly areas and critical assets?

SENSORS AND TELEMETRY

Proposed evidence. Proposed system interfaces preserve original source labels; fixed location tags can provide anonymous confirmation of selected physical labels. Coverage. Operationally relevant entrances, levels, stairs, fire and alarm zones, sectors, assembly areas, plant and life-safety assets. Timing, aggregation and privacy. Crosswalk versions at retrieval time; updates on every command sector change, panel reconfiguration or space renaming. Reference-name level only; no person, device or movement data. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. One-to-many mappings, legacy aliases and locally invented terms cannot always be reconciled automatically, especially when physical boundaries differ. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command owns operational names, while Estates space data, fire-safety management, alarm and system owners, Security Control and site liaison own source labels. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the current incident sector plan, room and floor register, entrance and assembly schedule, alarm-zone map, asset register and direct label observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a source predates renaming or reconfiguration, command has changed a sector boundary, or two current authoritative records conflict. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison - multi-source terminology crosswalk and ambiguity detection. Decision interval. at initial handover, whenever a new sector or resource point is named and after any reported misunderstanding. Minimum evidence. the exact source labels, current authoritative registers, command-approved operational names and explicit unresolved relationships. Method. display original terms side by side, map only verified equivalences, show one-to-many relationships and preserve command terminology without silently rewriting source logs. Decision discipline. Keep conflicts visible. Uncertainty. state version, boundary mismatch, historical-only labels and terms that cannot be uniquely mapped. Safe failure. show NAMING CONFLICT and require an explicit command-agreed reference.

ANSWER BOUNDARY

Do not rename command sectors, infer equivalence from similar text, overwrite source records or let the May 2021 floor plan establish current names or boundaries. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Reduces avoidable radio and handover errors caused by agencies referring to the same place or asset with different labels.

STAKEHOLDER QUESTION

ER-057

Which emergency services and specialist capabilities are reported on scene, en route, requested or unavailable, and where is the agreed liaison point?

SENSORS AND TELEMETRY

Proposed evidence. Agency, capability and estimated-arrival status must come from authorised emergency-service reports and command logs. A role-coded site acknowledgement may show only that a liaison function has checked in; no responder tracking, vehicle registration capture or personal-device location is required. Coverage. Agencies and specialist capabilities relevant to the incident, plus the agreed multi-agency liaison point. Timing, aggregation and privacy. Event-driven status updates with source and retrieval times; report capability and role, not individual movement or roster details beyond operational need. Quality and provenance. Retain reporting agency, command reference, update authorisation, clock offset, latency, superseded statuses and unresolved duplicates. Limits. An arrival acknowledgement can lag; a vehicle or agency listing does not prove that a capability is ready, tasked or available to command. Deployment status. Proposed; show STATUS NOT CONFIRMED unless an authorised source and current update are available.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service control, followed by multi-agency liaison officers and the authorised site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current attendance and mobilisation records, command log, specialist resource requests, liaison-point decision and direct agency check-in. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when status has not been reconfirmed within the command reassessment interval, a resource has been reassigned or released, or the liaison point has changed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + observation - agency and capability status reconciliation. Decision interval. At mobilisation updates, first arrival, multi-agency handover and whenever a requested capability changes. Minimum evidence. An authorised command or control report, agency or capability label, status time and agreed liaison point. Method. Reconcile reported, requested, en-route, on-scene and unavailable states; keep capability separate from vehicle or person count and retain superseded reports. Decision discipline. Presence does not equal readiness or tasking. Uncertainty. Show reporting delay, ambiguous capability labels and conflicting agency reports. Safe failure. Display CAPABILITY STATUS NOT CONFIRMED and leave tasking to incident command.

ANSWER BOUNDARY

Do not identify or track responders, infer readiness from vehicle presence, authorise tasking or expose personal roster data beyond operational need.

WHY THIS IS USEFUL

Supports orderly multi-agency liaison and stops command assuming a capability is ready merely because an agency or vehicle is listed.

STAKEHOLDER QUESTION

ER-058

If the command post must move, which pre-defined alternatives have current evidence for access, separation, communications, power and space?

SENSORS AND TELEMETRY

Proposed evidence. Proposed external smoke and weather instruments, anonymous area counts, lighting and power health, gate and route status, noise sensors and communications-infrastructure checks can describe nominated sites. Coverage. Pre-defined command-post and rendezvous candidates with their immediate access, evacuation and casualty-route interfaces. Timing, aggregation and privacy. Environmental and occupancy summaries every minute, access and power changes as events and communications health every 30 seconds. Candidate-location conditions only; no person identification, conversations or device tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timeline and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Point sensors may not represent the whole area, conditions can change quickly and good infrastructure indicators do not establish tactical security, separation or communications. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, police and ambulance commanders where relevant, Security Control, Estates power and communications owners and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current command requirements, direct inspection, emergency site plan, live environmental and infrastructure data, access and crowd reports and temporary closure information. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when environmental or crowd data is older than 2 minutes, a route or power state is older than 5 minutes, or command requirements have changed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Observation + comparison + recommendation (options only) - command-post relocation comparison. Decision interval. When the current post is compromised, communications or access deteriorate, the cordon changes or command requests relocation options. Minimum evidence. At least two pre-defined alternatives, current access and separation evidence, communications status, basic power or lighting information and command-defined constraints. Method. Compare each candidate against the same criteria and expose unobserved or conflicting factors; do not rank tactical security or select the post. Decision discipline. Options only; command decides and direct inspection confirms. Uncertainty. Show sensor distance, age, coverage gaps and any condition likely to change quickly. Safe failure. Display NO RELOCATION POINT VERIFIED and provide only the underlying evidence.

ANSWER BOUNDARY

Do not select or authorise a command post, predict plume or crowd movement, assess tactical security or present a low-risk score as proof of suitability.

WHY THIS IS USEFUL

Lets incident command compare relocation options when the current post becomes compromised without treating telemetry as a tactical selection.

STAKEHOLDER QUESTION

ER-059

Which critical facts changed since the last briefing, who owns each update, and which conflicts remain unresolved?

SENSORS AND TELEMETRY

Proposed evidence. Proposed event interfaces can contribute time-stamped system and environmental changes; manual operational facts require an authorised role, observed time and source attribution. Coverage. Incident location, access, hazards, occupancy, life-safety systems, isolations, communications, staging and recovery restrictions. Timing, aggregation and privacy. Event-driven with a current-state view and immutable history; refresh at every command-defined reassessment and handover. Fact, zone or asset level; personal details excluded unless handled separately under emergency-service authority. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. The service cannot know unreported changes, and a latest timestamp is not automatically the most authoritative or accurate source. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational facts, with each site or system fact owned by the competent responsible function and the site liaison accountable for site handover. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current command and incident logs, direct crew reports, authoritative system panels, controlled records, approved manual updates and source-health data. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a fact has no owner, observed or effective time, has passed its reassessment interval, or a higher-precedence source contradicts it. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + diagnosis - provenance-aware fact history and unresolved-conflict register. Decision interval. continuously, with formal snapshots at command briefings, shift changes and handover. Minimum evidence. a defined fact, source, accountable owner, time, current state and explicit supersession or conflict relationship. Method. preserve all versions, apply source precedence without deleting lower-source evidence, highlight material changes and route unresolved conflicts to named owners. Decision discipline. Root cause needs competent confirmation; keep conflicts visible. Uncertainty. show whether the change is confirmed, reported, estimated, conflicting or unavailable and why resolution has not occurred. Safe failure. retain the conflict and display FACT NOT RESOLVED rather than select a convenient value.

ANSWER BOUNDARY

Do not resolve command disputes, fabricate missing facts, identify people, authorise actions or let the latest telemetry silently override direct observation or authoritative control. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Keeps briefings focused on what has materially changed and prevents stale or contradictory facts being carried forward as settled truth.

STAKEHOLDER QUESTION

ER-060

Which verified building information and local system views remain available when site networks, cloud services or normal power are unavailable?

SENSORS AND TELEMETRY

Proposed evidence. Proposed local-cache, gateway, UPS, network-path and offline-terminal health can report which pre-positioned information packages and local system views remain accessible. Coverage. Premises information box, offline emergency terminal, local fire and life-safety panels, approved printed packs and resilient site communications points. Timing, aggregation and privacy. Health every minute, package synchronisation and signature time on each update, and last successful offline access test. System and package availability; no user tracking beyond security-required access logs held by the authorised system owner. Quality and provenance. define health thresholds and test coverage. Record calibration applicability/due state, clock offset, heartbeat, uptime, latency, loss, gaps, saturation, range and mapping/configuration drift; retain raw health, device/firmware and transformation provenance. Limits. A healthy cache can contain stale data, batteries can deplete, local panels may be inaccessible and offline packages cannot contain live conditions after disconnection. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for reliance, then the fire-safety responsible person, Estates digital operations, each local system owner, Security Control and information governance. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current signed offline package manifest, premises-information-box inventory, local panel availability, UPS and gateway status, continuity test records and direct access confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the package signature is older than its content reassessment interval, a material change occurred after synchronisation, an offline test is overdue or local access is not directly confirmed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + diagnosis - resilient-information availability and staleness assessment. Decision interval. at initial handover, on network or power degradation and during continuity and prolonged-incident reassessments. Minimum evidence. a verified local access point, current package version and scope, integrity check, power or battery state and explicit list of information that will no longer update. Method. classify each source as live local, cached, printed or unavailable; show synchronisation time, content limitations and dependency on power, credentials or physical access. Decision discipline. Root cause needs competent confirmation. Uncertainty. state cache age, unsynchronised changes, expected battery duration, inaccessible locations and whether live panel data remains available. Safe failure. show OFFLINE INFORMATION NOT VERIFIED and direct crews to physical panels, command and competent site representatives.

ANSWER BOUNDARY

Do not promise continuity, bypass access controls, expose restricted data, treat cached information as current live state or allow offline data to override direct observation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Shows responders what can still be trusted locally during an outage and what information has stopped updating.

STAKEHOLDER QUESTION

ER-061

Which areas can stage appliances and specialist resources without blocking fire access, evacuation, casualty routes or public-protection operations?

SENSORS AND TELEMETRY

Proposed evidence. Proposed bay and road-segment occupancy sensors, gate state, surface-water and weather sensing, anonymous crowd counts and authorised obstruction reports can describe nominated staging areas. Coverage. Pre-defined external staging and holding areas, adjoining access lanes, evacuation exits, casualty handover routes and public cordon interfaces. Timing, aggregation and privacy. Occupancy and environmental summaries every minute, access changes as events and manual reports with observed and expiry times. Area and route-segment level; no vehicle registration, responder identity or persistent movement tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Sensors cannot establish pavement load capacity, tactical security, future crowd movement or safe vehicle manoeuvring, and uninstrumented obstructions can appear quickly. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, police and ambulance commanders where relevant, emergency-service logistics, Estates transport and Security Control. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current staging and access plan, fire access requirements, traffic and cordon decisions, surface and load records, live site reports and direct marshal or crew observation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a staging plan predates access change, a protected route has moved, area conditions are older than 2 minutes or occupancy state is not directly confirmed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + comparison + recommendation (options only) - staging-area constraint and conflict comparison. Decision interval. before committing resources, after cordon or casualty-route changes and during major reinforcement or relief rotations. Minimum evidence. command-defined protected routes, at least one named staging candidate, current access and occupancy evidence, known fixed constraints and direct confirmation before vehicle placement. Method. display candidate capacity assumptions, access, conflicts, environmental conditions and evidence gaps; never allocate positions or sequence vehicle movements. Decision discipline. Options only; command decides. Uncertainty. state unmeasured load and turning capacity, temporary obstruction risk, sensor coverage and conflict with marshal reports. Safe failure. show STAGING SUITABILITY NOT ESTABLISHED and leave positioning to command and on-scene marshals.

ANSWER BOUNDARY

Do not reserve or clear space, direct vehicle movement, certify loading or safety, identify vehicles or people, or override police, fire or ambulance command. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps logistics leaders avoid placing resources where they will block critical routes or have to be moved during escalation.

STAKEHOLDER QUESTION

ER-062

Where are the relevant hydrants, tanks, pumps or other firefighting water supplies, and what current condition evidence exists for each?

SENSORS AND TELEMETRY

Proposed evidence. Proposed pressure loggers on selected private mains, tank-level, pump-state and valve-tamper sensors can support site water information; most public hydrant condition will rely on water-authority and fire-service records. Coverage. Private hydrants, tanks, pumps and designated water points on the site, plus mapped public hydrants near agreed appliance approaches. Timing, aggregation and privacy. Private pressure and level every 30 seconds, state changes as events, and public records by inspection or update date. Water asset and supply-zone level; no appliance tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Static or remote pressure does not prove flow under demand, public network conditions can change, hydrants can be obstructed or damaged and site telemetry may cover only private assets. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and fire-service water officers, then the water undertaker for public mains, Estates water owner for private systems and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire-service and water-undertaker hydrant records, private water schematics, inspection and flow-test results, tank and pump status, impairment records and direct hydrant checks. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when inspection or flow test is overdue, a road or main change affects access, live private pressure is older than 1 minute or an asset location is unverified. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - water-source location and evidence-readiness retrieval. Decision interval. during initial water planning, before relying on an alternative source and after any pressure, pump, valve or access change. Minimum evidence. verified asset location and ownership, latest inspection or test, current access condition and direct fire-service confirmation for the source to be used. Method. rank no source tactically; map public and private assets, show recorded capacity or test evidence, current private indications and all limitations or access constraints. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state test age, public-network live-data absence, pressure-versus-flow limitation, obstruction risk and mapping confidence. Safe failure. return WATER-SOURCE CONDITION NOT VERIFIED and rely on fire-service and water-undertaker procedures.

ANSWER BOUNDARY

Do not operate hydrants, pumps or valves, certify pressure, flow or water quality, direct appliance positioning or promise supply availability. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Reduces search time for water assets while keeping actual usability subject to fire-service testing and the responsible water owner.

STAKEHOLDER QUESTION

ER-063

What does the designated fire-fighting lift indicate for recall, fire-service mode, power, car position, lobby conditions and communication?

SENSORS AND TELEMETRY

Proposed evidence. Proposed lift-controller state, car position, recall and fire-service mode, landing-door state, primary and secondary power, lobby smoke and pressure, and lift communication health can provide indications. Coverage. Each designated fire-fighting lift, machine or control space, power supply, protected lobby and associated communication point. Timing, aggregation and privacy. Lift and power changes as events, lobby conditions every 30 seconds and controller heartbeat every 15 to 30 seconds. Lift, landing and lobby level; no passenger, access-card or responder tracking. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. Controller state cannot prove shaft, car or machinery safety, lobby sensors cover limited points and fire, water or power damage can invalidate normal diagnostics. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and direct crews, then the lift responsible person, electrical authorised person, fire-safety management, Estates lift owner and maintenance contractor. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire strategy and lift designation, controller and power status, inspection and maintenance records, impairment log, lobby and communication state and direct competent confirmation. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the lift designation or fire strategy predates change, controller data is older than 30 seconds, maintenance or fault status is unresolved, or local observation conflicts. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + diagnosis - fire-fighting-lift dependency readiness assessment. Decision interval. at initial vertical-access planning, before reliance on the lift and after any power, recall, lobby, water or fault change. Minimum evidence. verified designation and served levels, current controller and fire-service mode, both required power conditions where applicable, lobby condition evidence and direct confirmation by competent personnel or crews. Method. display each dependency separately as confirmed, indicated, faulted or unknown; compare intended recall with actual controller events without recommending use. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unobserved shaft conditions, sensor locations, controller limitations, water ingress risk and whether local inspection has occurred. Safe failure. show FIRE-FIGHTING LIFT NOT VERIFIED FOR USE and require command and competent lift assessment.

ANSWER BOUNDARY

Do not call, recall, operate or reset the lift, authorise use, certify safety or protection, track occupants or let controller status override direct crew checks. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Provides a disciplined dependency check so command can see why an apparently running lift may still be unsuitable or unverified.

STAKEHOLDER QUESTION

ER-064

What is the indicated status of passenger, evacuation and service lifts, and what does the current fire strategy allow responders to rely on?

SENSORS AND TELEMETRY

Proposed evidence. Proposed lift-controller mode, position, door, fault, entrapment alarm, power and communication status can describe each lift installation. Coverage. All passenger, evacuation and service lifts in Abacws, with explicit designation and served levels. Timing, aggregation and privacy. Controller and alarm changes as events, heartbeat every 30 seconds and inspection or designation by effective date. Lift and landing level; no passenger identity or journey history. Quality and provenance. competent tests must confirm point mapping, cause-and-effect and interfaces. Use original controller time; record clock offset/latency. Show heartbeat, power, network, fault, isolation, test, missed polls and configuration drift; retain source events unchanged. Limits. A lift shown parked or healthy may have unsafe shaft, lobby, power or water conditions, and a normal passenger lift is not made suitable by availability alone. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command, the lift responsible person, fire-safety management, Estates lift and electrical owners and maintenance contractor. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current fire strategy and evacuation-lift plan, lift designation and served-level schedule, controller state, faults and entrapment reports, maintenance records and direct competent observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the strategy predates alteration, designation is unclear, controller data is older than 30 seconds, or a fault or water ingress report is unresolved. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - lift-type, intended-role and current-state reconciliation. Decision interval. at initial life-safety handover, on any lift or power event and before any recovery or accessibility planning relies on a lift. Minimum evidence. current designation and strategy, controller health and mode, power and communication state, relevant lobby or access conditions and a competent owner. Method. group lifts by intended incident role, show current indication and dependencies, and flag any conflict between controller labels, strategy and physical signage. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state unobserved car and shaft condition, water and smoke exposure, sensor limits and whether direct inspection occurred. Safe failure. show LIFT ROLE OR AVAILABILITY NOT ESTABLISHED and refer decisions to incident command and competent lift owners.

ANSWER BOUNDARY

Do not recommend or authorise lift use, operate or reset lifts, certify evacuation or fire-fighting suitability, identify passengers or infer entrapment outcomes. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents a lift that merely appears operational being mistaken for a lift intended and verified for emergency use.

STAKEHOLDER QUESTION

ER-065

Where are site emergency equipment caches, and which items were physically inspected, within date and confirmed serviceable?

SENSORS AND TELEMETRY

Proposed evidence. Cabinet open, seal or tamper state and local environmental alarms may indicate whether a site emergency cache has been accessed or exposed to damaging conditions. Item availability, expiry and serviceability must come from a signed physical inventory and competent inspection; do not rely on radio-tag counts or weight as proof of usable equipment. Coverage. Named, mapped emergency caches that the site owner has authorised for responder information; ordinary stores and private equipment are excluded. Timing, aggregation and privacy. State changes as events; inventory after each authorised opening and at scheduled checks. Item category, quantity range, last inspection and expiry only; no user tracking. Quality and provenance. Function-test contacts and alarms; retain cache ID, seal number where used, signed checker, inspection version, source-event and retrieval times, clock offset, uptime and gaps. Limits. A closed or sealed cabinet cannot prove equipment is complete, compatible, within date or serviceable, and items may be removed after an incident starts. Deployment status. Proposed; return CACHE NOT VERIFIED unless physical inventory, mapping, permission and current condition are confirmed.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational use, then the equipment owner, Security Control, Estates operations, Health and Safety and the site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current cache register, signed inspection and expiry records, issue and replenishment logs, condition checks, location plan and direct physical verification. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when inventory predates an opening or use, inspection or expiry is overdue, the cache location changed, or sensor indication conflicts with a manual check. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + observation + comparison - cache location and physical inventory status. Decision interval. Before relying on a site cache, after an authorised opening and at recovery handback. Minimum evidence. A mapped cache, current signed inventory, competent inspection or serviceability record, and direct confirmation of access where practicable. Method. Compare required item category with inspected quantity, expiry and serviceability; separate cabinet state from item condition. Decision discipline. Site equipment is supplementary and command decides whether it is acceptable. Uncertainty. Show inspection age, missing items, compatibility unknowns and any access or environmental alarm. Safe failure. Display CACHE NOT VERIFIED and do not present the equipment as available for use.

ANSWER BOUNDARY

Do not recommend or authorise equipment use, certify serviceability, infer contents from cabinet state or disclose security-sensitive cache details outside approved roles.

WHY THIS IS USEFUL

Helps responders decide whether a nearby site cache is worth physically checking without relying on unverified or incompatible equipment.

STAKEHOLDER QUESTION

ER-066

Which casualty collection or triage-area candidates have usable access, space, utilities, shelter and separation evidence for ambulance command to assess?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous area occupancy, lighting, power, temperature, air quality, weather, surface-water and access-route status can describe pre-defined candidate areas. Coverage. External and selected internal candidates with ambulance access, evacuation-route interfaces and utility points documented. Timing, aggregation and privacy. Environmental and occupancy summaries every minute, access and power changes as events and manual condition reports with observed time. Area condition and approximate capacity range; no patient identity, clinical monitoring or health-data processing. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Sensors cannot establish clinical suitability, infection control, privacy, security, future demand or the condition of individual casualties. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. ambulance incident command first, then fire and police command where relevant, Estates facilities and utilities owners, Security Control and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. ambulance command requirements, emergency site plan, direct inspection, access and utility status, weather and environmental readings, crowd and cordon information. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when conditions are older than 2 minutes, access or cordon arrangements changed, utility status is unverified or the candidate has not been directly inspected. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + comparison + recommendation (options only) - candidate-area evidence comparison for ambulance assessment. Decision interval. before selection, after casualty numbers or incident footprint changes and whenever access, weather, utilities or public separation changes. Minimum evidence. at least two named candidates, ambulance-defined requirements, current access and separation status, direct inspection and explicit utility and environmental limitations. Method. compare each candidate using the same evidence checklist, show capacity assumptions and route conflicts, and leave clinical and operational selection to ambulance command. Decision discipline. Options only; command decides. Uncertainty. state point-sensor coverage, unmeasured privacy and infection-control factors, changing demand and any condition known only from a report. Safe failure. show TRIAGE-AREA SUITABILITY NOT ESTABLISHED and require ambulance-command assessment.

ANSWER BOUNDARY

Do not select or authorise a triage area, allocate patients, infer clinical need or survival, provide medical advice, identify people or promise utilities and separation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives ambulance command practical site evidence for comparing locations without encroaching on clinical or command decisions.

STAKEHOLDER QUESTION

ER-067

Which decontamination-area candidates have controlled access, water, drainage information, lighting and separation between dirty and clean routes?

SENSORS AND TELEMETRY

Proposed evidence. Proposed wind and weather, water-supply status, drain level or flow, surface slope and water accumulation, lighting and anonymous area occupancy can describe pre-defined candidate zones. Coverage. External hard-standing and service areas with mapped drains, water points, access controls and separation from public, clean and casualty routes. Timing, aggregation and privacy. Environmental and water conditions every minute, access changes as events and drainage or containment observations with time. Area and drain-segment level; no person identification, exposure estimation or tracking through decontamination. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Drain maps may be wrong, runoff can bypass sensors, wind shifts, contamination characteristics are unknown and fixed data cannot establish hazmat suitability. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and hazardous-material or ambulance decontamination leads, then Estates water and drainage owners, environmental compliance, Security Control and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. specialist command requirements, current drainage and water plans, environmental and spill controls, direct area inspection, access and cordon decisions and live conditions. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when drainage or water records predate works, conditions are older than 2 minutes, a cordon changed or the candidate has not been directly inspected. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + comparison + recommendation (options only) - decontamination-candidate constraint and runoff-pathway comparison. Decision interval. when decontamination is being considered, after wind, access, water or contamination changes and before recovery handback. Minimum evidence. specialist-defined requirements, a named candidate, current water and access status, drainage ownership and pathway information, direct inspection and known contamination limitations. Method. show water, access, slope, drainage connections, separation and evidence gaps; do not rank tactical preference or model contaminant fate beyond bounded pathways. Decision discipline. Options only; command decides. Uncertainty. state drain and subsurface unknowns, sensor specificity, wind variability, contamination uncertainty and unobserved runoff routes. Safe failure. show DECONTAMINATION AREA NOT VERIFIED and require specialist command and environmental assessment.

ANSWER BOUNDARY

Do not select or authorise decontamination, estimate exposure, direct runoff or treatment, operate valves or drains, identify people or certify environmental containment. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps specialist leads avoid a location whose drainage, access or clean-route conflicts are only discovered after operations begin.

STAKEHOLDER QUESTION

ER-068

Which responder welfare, rest and rehabilitation spaces and basic services are available for a prolonged incident, and what restrictions apply?

SENSORS AND TELEMETRY

Proposed evidence. Proposed anonymous occupancy, temperature, humidity, air quality, lighting, power, water and toilet-service status can describe nominated welfare areas. Coverage. Pre-defined internal and external responder welfare, briefing and rest candidates outside operational and public routes. Timing, aggregation and privacy. Environmental and occupancy summaries every 5 minutes, utility changes as events and cleaning or closure status by effective time. Area condition and capacity range; no biometric, fatigue, health, attendance or individual usage monitoring. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Building conditions cannot determine individual fitness, hydration or rehabilitation need, and a room may be physically available but operationally unsuitable or contaminated. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service welfare leads, then Estates facilities, cleaning and utilities owners, Security Control and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current welfare plan, direct area inspection, utility and environmental status, cleaning and access restrictions, contamination or cordon information and accountable site reports. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when conditions are older than 10 minutes, access or contamination status changed, service availability is unverified or direct inspection has not occurred. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + comparison + recommendation (options only) - welfare-area availability and basic-service summary. Decision interval. during prolonged operations, relief planning, major weather change and each shift handover. Minimum evidence. a named area outside restricted zones, direct access confirmation, current utility and environmental status and command approval for operational use. Method. display space, access, temperature, air, lighting, water, toilets and restrictions; show capacity as a planning range and exclude individual health assessment. Decision discipline. Options only; command decides. Uncertainty. state sensor coverage, contamination unknowns, service duration, cleaning status and whether the area has been physically inspected. Safe failure. show WELFARE AREA NOT CONFIRMED and use emergency-service welfare arrangements.

ANSWER BOUNDARY

Do not assess responder fitness, fatigue or health, identify users, allocate rest, certify hygiene or air safety, or authorise a welfare location. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Supports sustainable incident staffing by showing where basic site facilities may exist and what still needs command and physical verification.

STAKEHOLDER QUESTION

ER-069

What site support can reduce the movement burden for breathing-apparatus cylinders, heavy equipment and repeated trips between staging and entry?

SENSORS AND TELEMETRY

Proposed evidence. Door, lift, loading-bay, route-lighting and temporary-power status may describe movement dependencies. Availability of trolleys, hoists, cylinder-handling points or other site support must come from a signed physical inventory and competent owner; the service does not assess emergency-service equipment. Coverage. The nominated staging-to-entry routes, service lifts or goods routes, approved support points and any authorised site cache. Timing, aggregation and privacy. Asset state changes as events, route-condition reports with observed and expiry times, and inventory at each authorised opening. No responder or equipment tracking. Quality and provenance. Function-test monitored states; retain asset ID, inspection or inventory version, clock offset, latency, uptime, gaps, mapping changes and source references. Limits. Site equipment may not meet emergency-service standards; lift, route and demand conditions can change, and telemetry cannot determine handling risk or cylinder serviceability. Deployment status. Proposed; show SITE SUPPORT NOT VERIFIED unless current physical confirmation and competent ownership are available.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service logistics or BA control, then Estates facilities and lift owners, Security Control and equipment custodians. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. command logistics plan, current access and lift status, verified dimensions and load information, site equipment inventory and inspection, direct route check and authorised site support offers. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an inventory predates use, route or lift state is older than 2 minutes, dimensions are unverified or the support offer lacks an accountable owner. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + recommendation (options only) - logistics-dependency and site-support screening. Decision interval. during BA or heavy-equipment planning, on reinforcement and relief cycles and after any route, lift or staging change. Minimum evidence. command-defined logistics need, named origin and destination, current route and lift evidence, verified site support item and direct suitability check by the relevant service. Method. map handling routes and dependencies, show site equipment condition and restrictions, and leave quantities, cycles and deployment decisions to emergency-service control. Decision discipline. Options only; command decides. Uncertainty. state load, width and serviceability unknowns, unmonitored segments, inventory age and rapidly changing demand. Safe failure. show SITE LOGISTICS SUPPORT NOT VERIFIED and rely on emergency-service equipment and procedures.

ANSWER BOUNDARY

Do not allocate cylinders or equipment, direct BA operations, certify lifting or load capacity, identify crews, operate lifts or promise site resources. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Reveals whether the building can reduce avoidable carrying distance or provide support without assuming site assets are fit for emergency-service use.

STAKEHOLDER QUESTION

ER-070

Which lighting, temporary-power, generator, distribution and cable-route constraints could disrupt night-time or prolonged operations?

SENSORS AND TELEMETRY

Proposed evidence. Proposed fixed illuminance points, distribution-board, generator, UPS and temporary-power status, and selected monitored distribution assemblies may provide evidence; cable routes and trip hazards require direct or authorised inspection. Coverage. External command and staging areas, principal responder routes, service yards and approved temporary-power connection points. Timing, aggregation and privacy. Power changes as events, lighting every minute, generator and fuel status every minute and temporary arrangements by effective and inspection time. Area, circuit and route-segment level; no person or equipment-user tracking. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. Point illuminance does not prove visibility, cables can be moved or damaged, capacity and protection require electrical assessment and generators may fail or run out of fuel. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and emergency-service logistics, then the electrical authorised person, Estates energy owner, temporary-power contractor and site liaison. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current electrical distribution and approved connection information, generator and UPS records, temporary installation certificates and inspections, cable-route plan, lighting status and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a temporary arrangement lacks current inspection, power data is older than 1 minute, route layout changed or electrical capacity and protection are not confirmed. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - temporary-lighting and power dependency assessment. Decision interval. before establishing temporary power or lighting, after any trip, relocation or weather change and throughout prolonged operations. Minimum evidence. defined operational load and area, an approved connection or independent supply, current source and protection status, inspected cable route and competent electrical owner. Method. show source, capacity record, protection, fuel or battery dependency, route conflicts and lighting measurements; do not design or authorise an installation. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state unmeasured visibility, load diversity, cable damage, generator duration, sensor location and inspection scope. Safe failure. show TEMPORARY POWER OR LIGHTING NOT VERIFIED and require emergency-service and competent electrical arrangements.

ANSWER BOUNDARY

Do not connect, energise or isolate circuits, design temporary power, certify capacity or safety, direct cable placement or promise lighting duration. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps logistics teams identify power and cable constraints before a prolonged operation is disrupted by trips, darkness or blocked routes.

STAKEHOLDER QUESTION

ER-071

Which alarm, sensor, system and incident-log records are at risk of overwrite or loss and require authorised preservation with provenance now?

SENSORS AND TELEMETRY

Proposed evidence. Proposed read-only export services can preserve original controller logs, telemetry packets, health records and configuration snapshots with cryptographic hashes; manual incident records require controlled capture. Coverage. All systems and records materially relevant to the incident, limited by legal authority, operational need, retention policy and technical availability. Timing, aggregation and privacy. Preserve source-event, export, receipt and retrieval times; capture volatile logs as soon as authorised without altering live control systems. System and record-set level; personal or video data remains segregated and subject to separate authorised handling. Quality and provenance. validate read-only capture, source time base and export completeness; logs need no calibration. Align source clocks to the incident timebase and show offset and latency; retain export/receipt times, uptime, gaps, overwrite risk, transformations, format/version and hashes; preserve originals and custody provenance. Limits. Logs may be incomplete, overwritten, clock-skewed or altered by normal system operation; a hash proves file integrity after capture, not truth or completeness of the source. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command and the lead investigating authority, then each system owner, university information governance, Security Control and the site incident-record custodian. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. formal preservation request, controller and gateway logs, configuration and clock records, incident and command logs, access and change records, export manifests and signed chain-of-custody forms. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when volatile data is approaching overwrite, a source has continued changing without a snapshot, export lacks original format or metadata, or custody and permission are unclear. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + comparison + recommendation (options only) - authorised evidence-preservation inventory and integrity logging. Decision interval. as soon as operationally safe and authorised, after each material system change, at handover and before system reset, remedial work or decommissioning. Minimum evidence. legal or command authority, identified source and owner, preservation scope, original time basis, read-only export method, hash and documented custodian transfer. Method. prioritise volatile sources, preserve originals and metadata, generate a manifest and hash, record every transfer and keep interpretation separate from acquisition. Decision discipline. Options only; command decides. Uncertainty. state missing intervals, clock quality, export transformations, source health, overwrite risk and whether records are original, copied or reconstructed. Safe failure. record EVIDENCE NOT PRESERVED OR AUTHORITY NOT CONFIRMED and avoid unauthorised acquisition or alteration.

ANSWER BOUNDARY

Do not seize, alter, reset or operate systems, bypass permissions, collect unrelated personal data, certify evidential admissibility or infer cause from preserved records. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Protects perishable digital evidence and gives investigators a transparent record of what was captured, when and by whom.

STAKEHOLDER QUESTION

ER-072

When and where did sensors, controllers, networks, clocks or data feeds fail, drift, saturate, lose coverage or become unreliable during the incident?

SENSORS AND TELEMETRY

Proposed evidence. Proposed health telemetry can record heartbeat loss, power state, packet loss, latency, clock offset, gateway restart, sensor saturation, calibration state and mapping failures. Coverage. Every commissioned feed used or considered for the responder service, including local devices, controllers, gateways, networks and cloud or offline components. Timing, aggregation and privacy. Health at 30-to 60-second intervals and on state change; preserve the last healthy and first failed times with source and retrieval times. Asset, interface and coverage-zone level; no user or communication-content monitoring. Quality and provenance. define health thresholds and test coverage. Record calibration applicability/due state, clock offset, heartbeat, uptime, latency, loss, gaps, saturation, range and mapping/configuration drift; retain raw health, device/firmware and transformation provenance. Limits. A missing heartbeat does not identify the cause, local logs may be lost with power and apparent recovery can leave corrupted, stale or mis-timed data. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational consequence, then each competent system owner, Estates digital operations, network operations and information governance. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. device and controller health logs, power and network events, clock-synchronisation records, configuration changes, maintenance and incident reports and direct panel checks. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when health records are missing or extracted after overwrite, a recovered feed has not been revalidated, or the system owner cannot confirm the mapping and time basis. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. observation + diagnosis - failure timeline and coverage-impact reconstruction. Decision interval. during any live data anomaly, at command handover and during post-incident technical reassessment. Minimum evidence. asset identity, known expected heartbeat, last healthy evidence, first failure or degradation evidence, time basis and affected coverage or outputs. Method. build a component dependency timeline, distinguish device, controller, network, clock and mapping failures, and show which displayed answers were affected at each time. Decision discipline. Root cause needs competent confirmation. Uncertainty. state unknown cause, missing logs, recovery validation, clock overlap and whether failure was total, intermittent or quality-related. Safe failure. mark affected evidence UNRELIABLE for the relevant interval and do not reconstruct unsupported values.

ANSWER BOUNDARY

Do not assign blame, infer cyberattack or sabotage without competent investigation, inspect network content, identify users, or treat restored connectivity as proof of data validity. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Shows responders and investigators exactly when a dashboard or feed should not be trusted and which decisions may need corroboration from other evidence.

STAKEHOLDER QUESTION

ER-073

What defensible event sequence can be reconstructed across alarms, building systems, calls, observations, reports and command actions?

SENSORS AND TELEMETRY

Proposed evidence. Proposed read-only event exports can provide original device, controller and gateway times; authorised manual records add observed and recorded times without modifying the source text. Coverage. Incident-relevant alarms, suppression, smoke control, lifts, access, power, environmental feeds, emergency calls, site reports and command log entries. Timing, aggregation and privacy. Preserve every available time basis and known clock offset; display events as ranges when ordering is not defensible. Event and source level; personal data minimised and segregated under investigating-authority permissions. Quality and provenance. validate read-only capture, source time base and export completeness; logs need no calibration. Align source clocks to the incident timebase and show offset and latency; retain export/receipt times, uptime, gaps, overwrite risk, transformations, format/version and hashes; preserve originals and custody provenance. Limits. Clocks drift, calls and reports are delayed, logs can be incomplete and sequence alone cannot establish causation, intent or the physical order of unrecorded events. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. the lead investigating authority and incident command, followed by each source-system owner, the site incident-record custodian and information governance. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. unaltered source logs, clock and time-zone configuration, communication and command records, manual reports, preservation manifests and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a log was transformed without provenance, clock offset is unknown, an event was entered retrospectively without disclosure or source coverage is incomplete. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + estimate - multi-source temporal reconciliation with interval ordering. Decision interval. during operational reassessment when chronology is needed and after evidence preservation for formal reconstruction. Minimum evidence. source-identifiable events, preserved original times, known or bounded clock uncertainty, provenance and explicit missing-source list. Method. normalise time zones, retain original timestamps, apply bounded offsets, group events whose intervals overlap and separate designed cause-and-effect from observed correlation. Decision discipline. Retain inputs, units and formula; label assumptions and range. Uncertainty. show time ranges, source latency, retrospective-entry risk, missing events and alternative valid orderings. Safe failure. state SEQUENCE PARTIAL OR ORDER UNRESOLVED and omit causal language. Forecast boundary: do not turn a trend or aligned sequence into a forecast without a verified model, horizon and error bound.

ANSWER BOUNDARY

Do not determine legal cause, intent, fault or compliance, identify individuals unnecessarily, alter evidence, or present time correlation as proof that one event caused another. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Creates a defensible shared timeline while preventing false precision from inconsistent clocks and incomplete records.

STAKEHOLDER QUESTION

ER-074

Which authorised fixed-area video or still-image records may be relevant, when will they be overwritten, and how can they be preserved without person-identification analytics?

SENSORS AND TELEMETRY

Proposed evidence. Camera-system health and recording-index metadata may expose camera ID, fixed field-of-view description, recording availability, clock quality and overwrite deadline. Talking Abacws must not stream, search or analyse image content. Coverage. Only authorised fixed-area cameras listed in the current camera register; blind spots, temporary cameras and unregistered changes remain gaps. Timing, aggregation and privacy. Event or segment metadata at the recording-system interval and an authorised preservation status; no facial, behavioural, gait or person-identification analytics. Quality and provenance. Verify camera mapping and clock offset, retain recorder ID, segment range, gaps, health, export hash, custody transfer, permissions and retrieval path. Limits. A recording may be obscured, mis-timed, unavailable or outside the field of view; metadata cannot prove image quality, relevance or evidential admissibility. Deployment status. Proposed; return AUTHORITY OR RECORDING NOT CONFIRMED unless lawful purpose, custodian, mapping and current availability are verified.

AUTHORITATIVE SUPPORTING DATA

Owners. the lead investigating authority, university data controller, Security Control, camera-system owner and information governance. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. lawful preservation request, current camera register and field-of-view description, recording index, retention policy, access record and chain-of-custody record. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when retention expiry is near, camera clock or location is unverified, recording health is unknown or permissions and custodianship are unresolved. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. Lookup + observation - recording availability and lawful preservation status. Decision interval. On authorised request, before overwrite and at every custody transfer. Minimum evidence. Lawful authority and purpose, relevant camera and time window, current recording availability, verified clock basis and named custodian. Method. Return metadata and preservation status only, keep media inside the controlled evidence system, hash authorised exports and record access. Decision discipline. Talking Abacws performs no image-content interpretation. Uncertainty. State field of view, clock accuracy, image-quality unknowns, recording gaps and whether preservation is requested, complete or unavailable. Safe failure. Display AUTHORITY OR RECORDING NOT CONFIRMED and do not access the media.

ANSWER BOUNDARY

Do not identify or track people, perform facial, behavioural or gait analysis, stream footage into the service, bypass access controls, disclose media or certify relevance or admissibility.

WHY THIS IS USEFUL

Preserves potentially relevant footage before overwrite while keeping media access, interpretation and personal-data handling inside authorised evidential controls.

STAKEHOLDER QUESTION

ER-075

Which isolations, overrides, temporary configurations or manual system changes were recorded during the response, and which may still remain active?

SENSORS AND TELEMETRY

Proposed evidence. Proposed switch, valve and controller feedback, role-coded command logs, temporary-configuration records and manual action entries can capture indicated changes without controlling equipment. Coverage. Life-safety, electrical, gas, water, ventilation, lifts, access control, alarms and other systems changed under incident arrangements. Timing, aggregation and privacy. Action, command, feedback and manual confirmation times preserved; reassessment and expiry times mandatory for temporary states. Asset and accountable role level; individual operator identities remain in authorised source systems where required. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. Manual actions may be missing, feedback can be wrong, local overrides can bypass central logs and a recorded command does not prove the physical state. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command until handover, then each competent system owner, the university duty manager and the recovery or facilities lead. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. command and action logs, permits and lock-off records, controller and event logs, current asset state, temporary-change register and direct competent verification. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a temporary action has no owner, effective or expiry time, physical state has not been rechecked, or command, feedback and observation disagree. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - change-state reconciliation and outstanding-action register. Decision interval. before every handover, after each authorised change and before system restoration or incident closure. Minimum evidence. asset identity, intended action, authority and owner, command or manual record, current feedback and direct verification where safety depends on state. Method. compare intended, commanded, indicated and physically confirmed states; keep unresolved changes open and link every temporary state to reassessment and expiry. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state missing actions, failed indicators, local overrides, unknown current state and whether the action remains within its authority period. Safe failure. treat the asset as STATE UNKNOWN and prevent automatic closure of the action.

ANSWER BOUNDARY

Do not operate or restore systems, authorise or validate isolations, identify individual operators unnecessarily, certify a state as safe or infer authority retrospectively. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents incident-created changes being forgotten and gives competent owners a precise verification and restoration checklist.

STAKEHOLDER QUESTION

ER-076

Which areas remain restricted after the incident, which authority owns each restriction, and what competent evidence is required before reoccupation?

SENSORS AND TELEMETRY

Proposed evidence. Proposed structural movement, temperature, humidity, air quality, moisture, power and life-safety health can support recovery observations where calibrated and within range. Coverage. Incident-affected rooms, floors, routes, plant areas and adjacent zones included in a formal restriction or handback plan. Timing, aggregation and privacy. Restrictions by effective and reassessment time; sensor observations at system-appropriate intervals with last calibration and health. Area and asset condition; no person monitoring or re-entry tracking beyond authorised access control. Quality and provenance. record position, range, accuracy, calibration/due date and drift checks. Align source clocks to the incident timebase; show offset and latency; retain interval, units and aggregation. Show uptime, gaps, saturation, out-of-range or relocation flags, plus device, firmware and gateway provenance. Limits. Normal telemetry cannot establish structural, electrical, contamination, fire-protection or occupational safety, especially after heat, water, smoke or physical damage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command until formal handover, then the building responsible person and competent structural, fire, electrical, environmental and health-and-safety professionals. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. formal restriction and cordon decisions, competent inspection and test reports, system impairment and restoration records, current risk assessment, authorised reoccupation decision and direct observations. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a restriction lacks scope, owner or reassessment time, an assessment predates deterioration or remedial work, required systems remain impaired or telemetry is outside calibration or coverage. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - restricted-area and evidence-requirement register. Decision interval. at incident handover, after each competent assessment or remedial work and before any proposed reoccupation phase. Minimum evidence. mapped restricted area, authority and effective time, required competent assessments, current evidence status and named decision owner. Method. show each restriction, outstanding requirement and evidence owner; distinguish monitoring observations from certification and retain restrictions until formal resolution. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state unseen damage, delayed deterioration, coverage gaps, conflicting assessments and conditions not addressed by sensors. Safe failure. retain the restriction and state REOCCUPATION NOT ESTABLISHED.

ANSWER BOUNDARY

Do not certify reoccupation, remove cordons, authorise access, operate systems or let normal sensor readings replace competent inspection and owner decisions. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives recovery leaders a traceable list of conditions and professional evidence needed before areas can return to use.

STAKEHOLDER QUESTION

ER-077

What restoration dependencies connect power, alarms, access, lifts, ventilation, water and the areas proposed for return to service?

SENSORS AND TELEMETRY

Proposed evidence. Proposed asset-state and health interfaces can show current availability and dependency nodes; the dependency model must come from verified controlled engineering information. Coverage. Incident-affected utility supplies, life-safety systems, access control, lifts, ventilation, water, drainage and occupied areas included in recovery planning. Timing, aggregation and privacy. Asset states on change, dependency records by controlled version and restoration actions with effective, verification and handover times. Asset, service and area level; no occupant tracking. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. Dependency records can omit manual bypasses, temporary supplies and hidden shared services; a restored component can fail under load or remain unsafe despite a healthy indication. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. the recovery lead and building responsible person, with each electrical, mechanical, lift, fire, access and water system owned by its competent professional. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current verified schematics and dependency register, restoration and test plans, permits, system impairment records, live states, commissioning evidence and competent handback approvals. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the dependency model predates modification, temporary arrangements are unrecorded, a restored state lacks test evidence or an upstream service remains unknown. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + calculation + comparison - dependency-graph validation and staged-restoration readiness display. Decision interval. during recovery planning, before each restoration stage and after any failed test or changed area priority. Minimum evidence. verified dependency links, current state of every prerequisite, competent owner, approved restoration sequence and required test or commissioning evidence. Method. trace upstream and shared dependencies, identify single points and temporary bypasses, show prerequisites and unresolved tests, and keep decision authority with competent owners. Decision discipline. Retain inputs, units and formula; keep conflicts visible. Uncertainty. state model completeness, hidden common services, temporary-state risk, indication-versus-test differences and downstream areas affected by failure. Safe failure. show RESTORATION DEPENDENCIES INCOMPLETE and prevent automatic readiness or closure status.

ANSWER BOUNDARY

Do not design or authorise restoration, energise or restart systems, certify service readiness, schedule reoccupation or let analytics override competent commissioning and handback. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Helps recovery teams avoid restoring an area whose alarms, ventilation, access or other essential dependencies are still unavailable or untested.

STAKEHOLDER QUESTION

ER-078

Which temporary controls and residual risks are authorised during recovery, who owns each one, and when must it be checked, renewed or removed?

SENSORS AND TELEMETRY

Proposed evidence. Proposed temporary alarm, environmental, access, power and barrier monitoring can provide verification signals where specifically commissioned for the control. Coverage. Each authorised temporary control, affected area and monitored condition identified in the recovery risk assessment. Timing, aggregation and privacy. Effective, inspection, reassessment and expiry times mandatory; sensor health and observations at the interval set by the competent control owner. Control, area and accountable role; no person monitoring beyond authorised access restrictions. Quality and provenance. static facts need dated survey accuracy/version; binary states need functional tests; measurements need calibration and drift checks. Align source clocks to the incident timebase and show offset and latency; retain heartbeat, uptime, gaps and mapping changes; retain raw event/sample and aggregation provenance. Limits. A monitoring signal cannot prove the control remains adequate, people may bypass arrangements and residual risks can change outside the sensor coverage. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. the building responsible person and recovery lead, with fire, structural, electrical, environmental, access and health-and-safety controls owned by competent professionals. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. current recovery risk assessment, authorised temporary-control register, permits, inspection and test records, monitoring status, exception approvals and handback decisions. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when a control has no owner or expiry, inspection is overdue, conditions changed, monitoring is unhealthy or the underlying permanent system has been altered. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. lookup + observation + comparison - temporary-control assurance and expiry register. Decision interval. at authorisation, every specified inspection, after a condition change and before expiry, extension or removal. Minimum evidence. defined residual risk, authorised control, scope, owner, effective and expiry times, check method and evidence of the latest check. Method. display controls against the risk they address, surface overdue or unhealthy verification, keep exceptions visible and require formal owner action for extension or closure. Decision discipline. Separate observation from interpretation; keep conflicts visible. Uncertainty. state monitoring coverage, human-dependency, unobserved failure modes, exception scope and whether the control has been physically checked. Safe failure. mark the control EXPIRED OR UNVERIFIED and retain the associated restriction until competent resolution.

ANSWER BOUNDARY

Do not authorise, extend or remove temporary controls, certify residual risk, operate systems, infer compliance from telemetry or permit reoccupation. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Prevents temporary safeguards becoming indefinite or silently ineffective during a long recovery period.

STAKEHOLDER QUESTION

ER-079

Where did controlled plans, registers, labels or telemetry disagree with responder observations or competent site findings?

SENSORS AND TELEMETRY

Proposed evidence. Proposed provenance logging can retain the exact telemetry, map version and system state shown at the time; authorised field observations can be recorded with location, time and role. Coverage. All incident-relevant locations, assets and information products for which a documented discrepancy was reported. Timing, aggregation and privacy. Record at first observation, preserve the displayed source state and revisit after investigation, correction and verification. Discrepancy, asset and area level; no performance scoring or attribution to individual workers. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. An observation can be incomplete, telemetry can fail transiently and a plan discrepancy may reflect temporary incident damage rather than a pre-existing record defect. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command for operational observations, then each data or system owner, Estates document control, fire-safety management and the recovery or assurance lead. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. crew and competent-person observations, photographs or notes under authorised evidential handling, displayed telemetry and health, controlled plan version, change history and correction record. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when the exact source version or displayed state was not preserved, the observation lacks time or location, or a correction has not been independently verified. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. comparison + diagnosis - source-to-field discrepancy register and resolution tracking. Decision interval. when a mismatch is first found, at handover and after corrective action or data update. Minimum evidence. specific observation, location or asset, observed time and role, exact conflicting source and version, operational impact and accountable resolver. Method. preserve both sides, classify mapping, state, timing, coverage or record-control failure, prioritise by safety impact and require independent verification before closure. Decision discipline. Root cause needs competent confirmation; keep conflicts visible. Uncertainty. state whether the discrepancy is incident-induced, intermittent, unverified or systemic and what evidence is still missing. Safe failure. keep DISCREPANCY UNRESOLVED and warn against relying on the affected source.

ANSWER BOUNDARY

Do not assign blame or competence ratings, alter original records, infer misconduct, identify individuals unnecessarily or close a discrepancy without accountable verification. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Turns responder observations into traceable corrections so the same unsafe map, asset label or telemetry error is not repeated at a later incident.

STAKEHOLDER QUESTION

ER-080

Which requested answers remain unavailable, conflicting or too uncertain, and which owner or evidence is needed to close the incident information pack?

SENSORS AND TELEMETRY

Proposed evidence. Proposed service-health and provenance data can show missing feeds, stale mappings, failed permissions and quality limits; no new fact is inferred merely to complete the pack. Coverage. Every catalogue question used during the incident and each evidence item included in command, handover or recovery outputs. Timing, aggregation and privacy. Maintain continuously and freeze an authorised snapshot at incident handover or closure, with later corrections versioned separately. Question, evidence source, asset or zone; personal information excluded unless held in a separate authorised record. Quality and provenance. records need no calibration; verify owner, version, effective period, location mapping and role. Record source clock offset, receipt latency and observed/entered times. Show gaps, superseded values and mapping drift with source-to-display provenance. Limits. Some questions are inherently unanswerable after evidence loss, and closure of an information gap does not prove operational or legal closure of the incident. Deployment status. Proposed; return UNAVAILABLE unless deployment, asset mapping, commissioning, permission and current health are independently verified.

AUTHORITATIVE SUPPORTING DATA

Owners. incident command until handover, then the lead investigating or recovery authority, supported by each accountable site, system and information owner. Precedence. Incident command, emergency-service information and direct observation first; then current competent site owners and authoritative panels; then commissioned, healthy and permissioned telemetry; then current controlled records. Sources. the complete question and answer log, source and permission register, unresolved-conflict list, preservation manifest, action owners, competent assessments and final handover record. Times, permission and provenance. show effective, observed, source-event and retrieved times, permission scope, owner, source/version and retrieval path. Currency. treat the answer as stale when an answer lacks current source, time or owner, an action has no due date, a conflict remains hidden, or later evidence has not been incorporated by versioned update. Conflicts. show values side by side, apply precedence and name the accountable owner needed to resolve them. Historical plan. The May 2021 floor plan is historical orientation only; it cannot establish current room use, access, services, hazards or closures.

ANALYSIS REQUIRED

Operation. comparison + diagnosis + recommendation (options only) - information-gap register and closure-readiness assessment. Decision interval. at command and shift handovers, before formal site handback and when compiling the final incident information pack. Minimum evidence. every requested answer classified as confirmed, reported, estimated, conflicting or unavailable, with source, time, limitation, owner and next evidence requirement. Method. enumerate unresolved questions, group by operational impact and responsible owner, record why evidence is insufficient and require explicit acceptance of any gap that cannot be closed. Decision discipline. Root cause needs competent confirmation; options only; command decides. Uncertainty. preserve unresolved states and explain whether the cause is missing evidence, permission, coverage, conflict, staleness or system failure. Safe failure. issue an INCOMPLETE INFORMATION PACK with open items rather than manufacture a complete narrative.

ANSWER BOUNDARY

Do not declare incident, investigation or recovery closure, fabricate missing evidence, certify completeness, assign blame or allow unresolved experimental telemetry to override authoritative judgement. Return unavailable or conflicting rather than infer a safe state.

WHY THIS IS USEFUL

Gives command, investigators and recovery leaders an honest handover of what is known, what is not and who must address each remaining evidence gap.

Catalogue coverage and decision map

The eight categories cover mobilisation, incident operations, life-safety, logistics and recovery without replacing command.

#	CATEGORY	QUESTION RANGE	PRIMARY RESPONDER DECISION	EVIDENCE BOUNDARY
1	Pre-arrival location, entrances and site access	ER-001 - ER-010	Confirm premises, approach, first entry, access resource, initial reporting point and temporary access change.	No driving command, gate operation, access authorisation, clearance guarantee or current-condition claim from the historical plan.
2	Incident location, routes and spatial orientation	ER-011 - ER-020	Reconcile incident location, route segments, naming, fallback routes, vertical access and specialist spaces.	No responder tracking, route-tenability promise or inference beyond mapped and directly observed segments.
3	Hazards, utilities, isolations and building information	ER-021 - ER-030	Find registered hazards and isolation references; expose persistent energy, structural and spread-path uncertainty.	No remote isolation, hazard certification, causal claim or substitution for competent engineering and specialist advice.
4	Occupancy, evacuation progress and inclusive access	ER-031 - ER-040	Present aggregate expectations, authorised evacuation or search declarations, inclusive assistance and accountability gaps.	No person identification, individual-survival prediction or declaration that a zone is clear from aggregate sensing.
5	Alarms, suppression, smoke control and life-safety systems	ER-041 - ER-050	Show alarm chronology, life-safety system indications, impairments and evidence-health gaps without controlling systems.	No acknowledgement, reset, disablement, control or certification of alarm, suppression, smoke-control or lift systems.
6	Communications, rendezvous, command and handover	ER-051 - ER-060	Maintain a shared command location, liaison, source-linked handover, communications picture and resilient local information.	No operational tasking, private contact disclosure, silent conflict resolution or generated fact without a source.
7	Staging, equipment, water, lifts and responder logistics	ER-061 - ER-070	Compare staging, water, lifts, site caches, casualty, decontamination, welfare and prolonged-incident logistics.	No equipment certification, clinical decision, decontamination authorisation or claim that a site asset meets responder standards.
8	Post-incident evidence, recovery and uncertainty	ER-071 - ER-080	Preserve evidence, expose data failures, reconstruct time, manage restrictions and dependencies, and retain unresolved uncertainty.	No blame assignment, evidential-admissibility claim, reoccupation approval or manufactured closure of missing information.

Across all categories: authoritative owners, direct observation and incident command take precedence; telemetry is proposed and must expose minimum evidence, uncertainty, permissions and safe failure.

Evidence and permissions rules

These rules apply to every question, data source, derived output, handover and recovery record in the catalogue.

1 COMMAND PRECEDENCE

Incident command, emergency-service information and direct observation take precedence over every record and telemetry item.

2 PROPOSED TECHNOLOGY

No sensor or integration is assumed to exist; deployment, mapping, commissioning, permission and current health must be verified.

3 HISTORICAL PLAN

The May 2021 floor plan is orientation only and cannot establish current room use, access, services, hazards or closures.

4 NO PERSON TRACKING

Do not identify, locate or follow people through faces, devices, access records, video or linked movement histories.

5 NO OPERATIONAL CONTROL

The service must not open doors, operate lifts, acknowledge alarms, isolate utilities, change settings or issue commands.

6 MINIMUM EVIDENCE

Every answer states the evidence needed; if it is absent, the permitted output is unavailable, conflicting or unconfirmed.

7 MEASUREMENT QUALITY

Name modality and location; show coverage, sampling, aggregation, calibration or test, uptime, gaps, saturation and drift.

8 TIME BASIS

Display effective, observed, source-event and retrieved times, plus clock offset, synchronisation source and latency.

9 SOURCE PROVENANCE

Retain owner, asset ID, source or document version, raw event or sample reference, transformations and retrieval path.

10 PERMISSION AND NEED

Use role-based access, purpose limitation, minimum necessary detail, retention controls and accountable custodians.

11 CONFLICTS STAY VISIBLE

Show contradictory current values side by side; apply precedence and name the competent owner needed to resolve them.

12 SAFE FAILURE

Do not manufacture certainty, safe states or complete narratives; state the gap and the evidence or owner required next.

Responder-facing outputs must remain concise, source-linked, permissioned and explicit about what is not known or not current.